Sustainable Materials, Green Fuels and Carbon Valorization:
the Next Generation of Biorefineries

Beyond Green: How Systems Modelling and Life Cycle Analysis Enable True Sustainability

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***Acknowledgement**: Project PID2023-151855OA-I00 funded by MICIU/AEI/10.13039/501100011033 and by FEDER, UE.*

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“Green” Does Not Always Mean Sustainable

Bio-based ≠ low carbon footprint

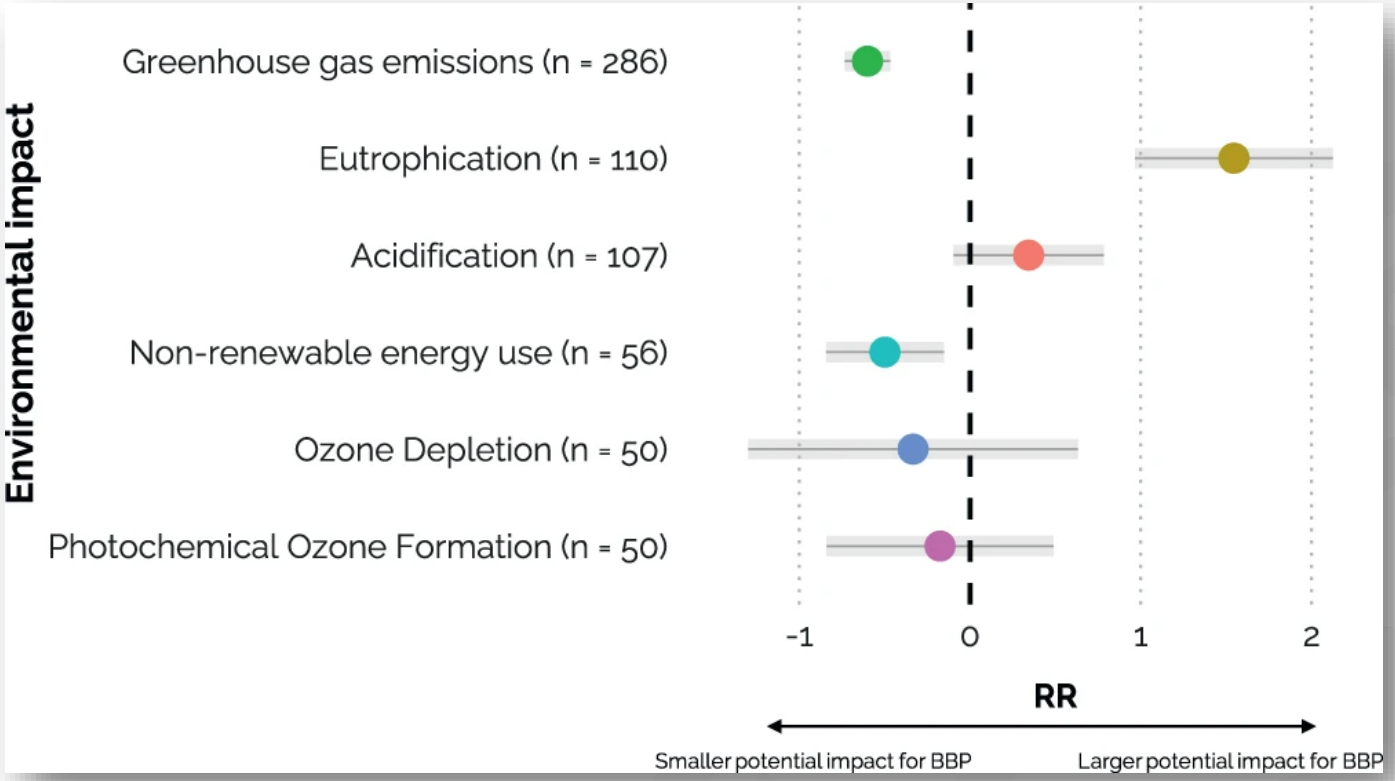
Recycling ≠ low impact

Carbon capture ≠ carbon neutral

Why Systems Modelling + LCA Matter?

Sustainability requires quantitative systems-level evaluation.

Motivation



Nat. Commun. 2023, **14**, 8521

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① Sustainable Materials

Biochar-Based Geopolymers from CDW.

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② Green Fuels

Multiproduct biorefinery from olive-derived biomass.

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① Sustainable Materials

Biochar-Based Geopolymers from CDW.

② Green Fuels

Multiproduct biorefinery from olive-derived biomass.

③ Carbon Valorization

Renewable carbon transition for the petrochemical industry.

1 Sustainable Materials

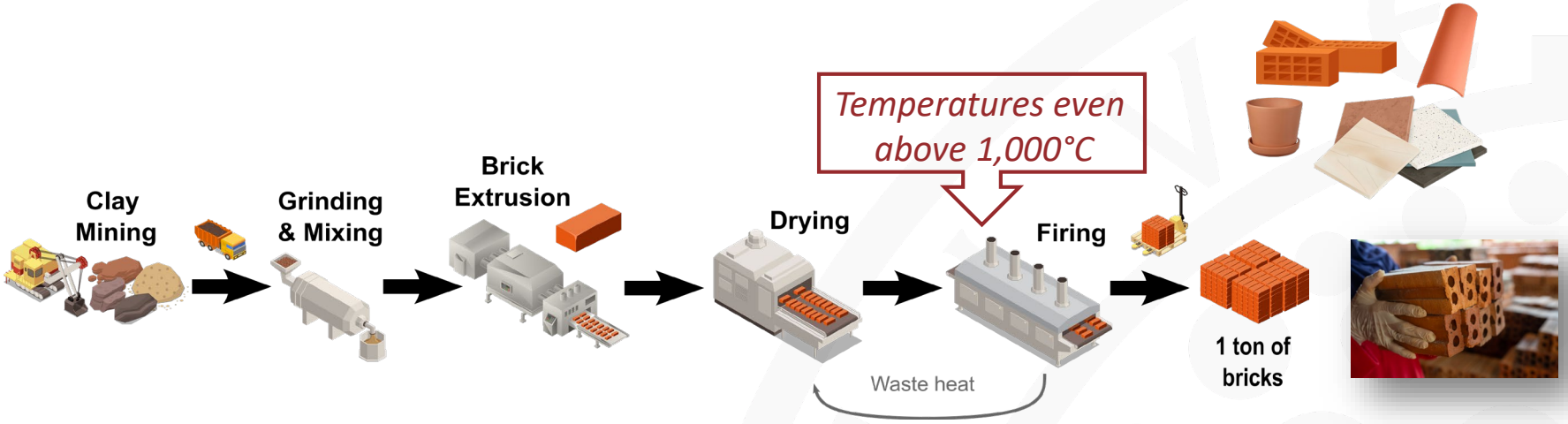
Biochar-Based Geopolymers from Construction and Demolition Waste: Experimental Study and Prospective Life Cycle Assessment under Net-Zero Scenarios



ACS Sustainable Resour. Manage. 2026, XXXX, XXX, XXX-XXX (*Under revision*)

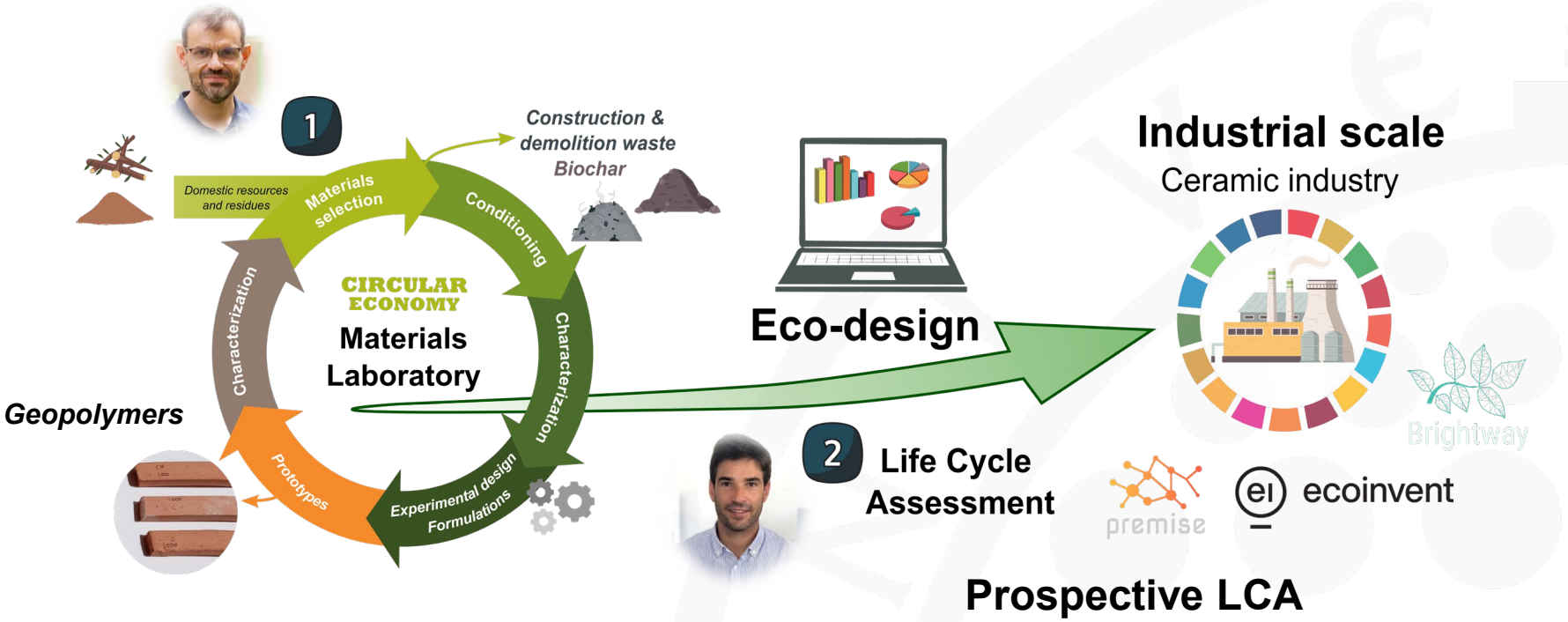
Sustainable Materials

- **Energy-intensive industries** emerge as **strategic players** to accelerate climate action and reach our sustainability goals.
- **Cement/Ceramic manufacturing are hard-to-abate industries**, given its intrinsic reliance on high-temperature operations and fossil-based thermal energy.



IDEA: To develop **novel geopolymers** that valorize **construction and demolition waste** and **biochar** (long-term carbon storage), with the aim of **reducing** the material **carbon footprint** and advancing toward **climate-neutral construction materials**.

Two levels of research: **experimental research** and life cycle assessment **modeling** to analyze the **technical feasibility and environmental performance** of novel biochar-based geopolymer materials.



*ECOCER – **ECO**design of **CER**amic materials for a sustainable construction.*

Experimental results

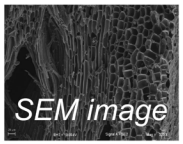
Raw materials

Construction & demolition waste  +  Chamotte ceramic waste

Biochar

Biochar addition *[0 - 7.50% mass]* 

- Elemental analysis
- Porosity and pH
- IR and SEM images

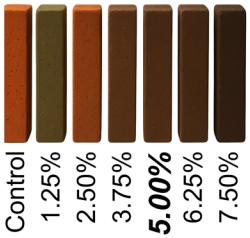


Activator

Sodium hydroxide (50%) + Sodium silicate (50%) + H_2O

Geopolymer prototypes

% Biochar added

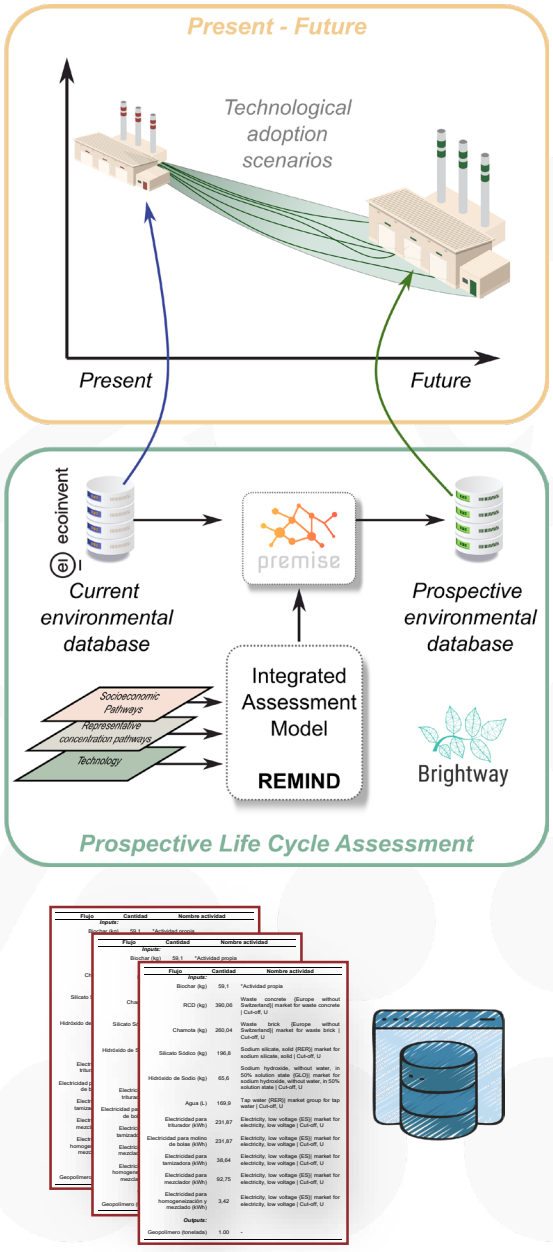


Optimun

Characterization of geopolymers (*84 specimens: 42 with a 7-day and 42 with a 28-day curing period*)

- Density, water absorption and porosity
- Mechanical strength in bending tests
- Compression Strength Test

25-30 MPa after 28 days



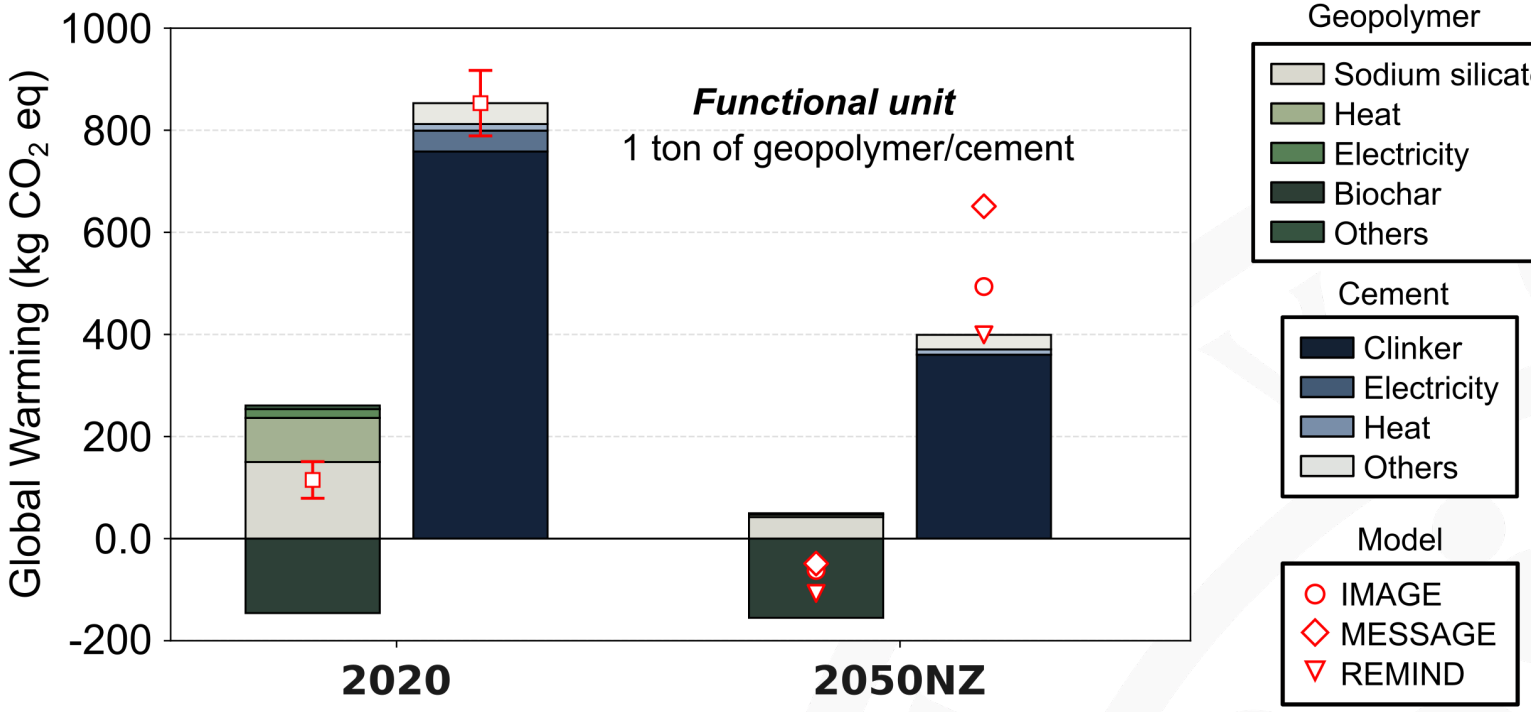
- ✓ Primary data from experiments to compile the life cycle inventory of the production system (foreground system).

Carbon footprint

Functional unit of 1 tonne of geopolymer

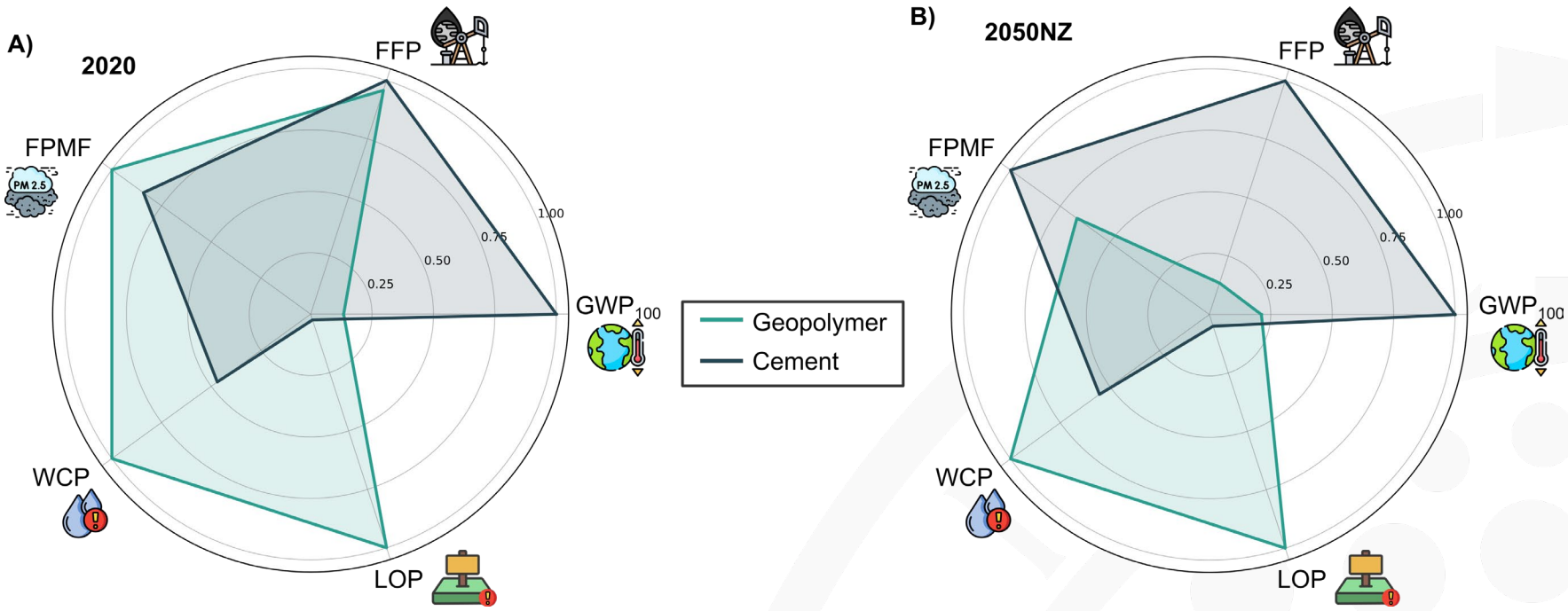
GWP100 (ReCiPe 2016)

Future **energy projections** using the **premise** toolbox.



- ✓ Geopolymer with 5% biochar **reduces** the **carbon footprint by ~70%** compared to conventional cement in 2020.
- ✓ In the **2050NZ scenario**, the geopolymer **achieves net-negative emissions (−107 kg CO_2 eq)**, while cement still emits ~402 kg CO_2 eq per ton.

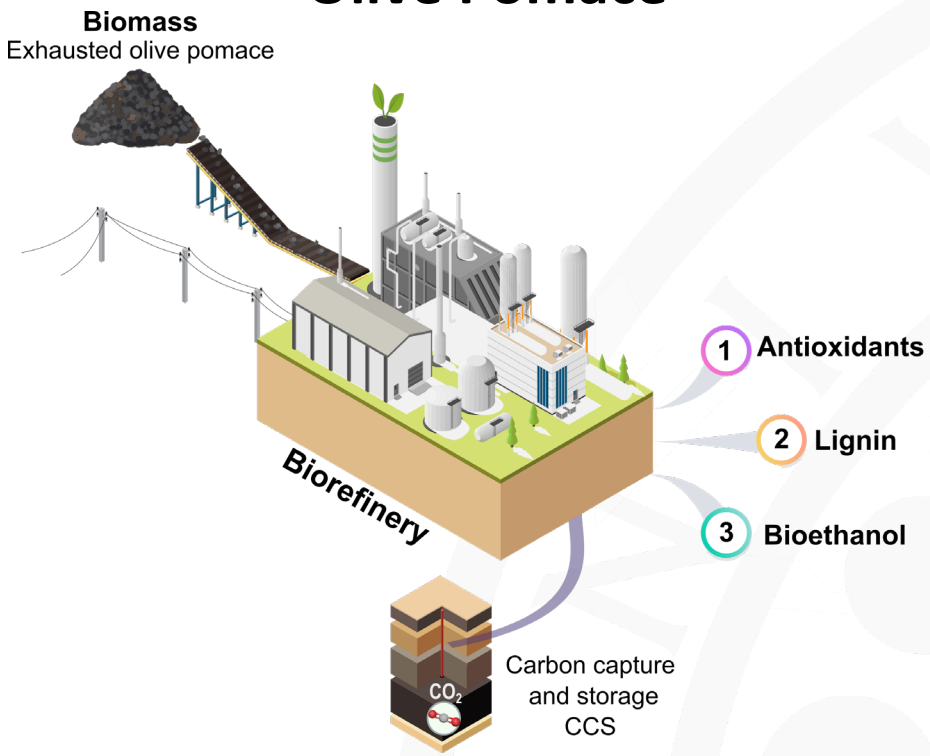
Broader Environmental Trade-offs



- ✓ Geopolymer showing **higher** relative **burdens in categories such as WCP and LOP**, particularly linked to resource and land-related impacts.
- ✓ **Strong climate-mitigation potential**, but its overall sustainability requires a **multi-impact life cycle perspective** beyond carbon footprint alone.

2 Green fuels

Techno-Economic and Environmental Assessment of a Multiproduct Biorefinery Based on Exhausted Olive Pomace

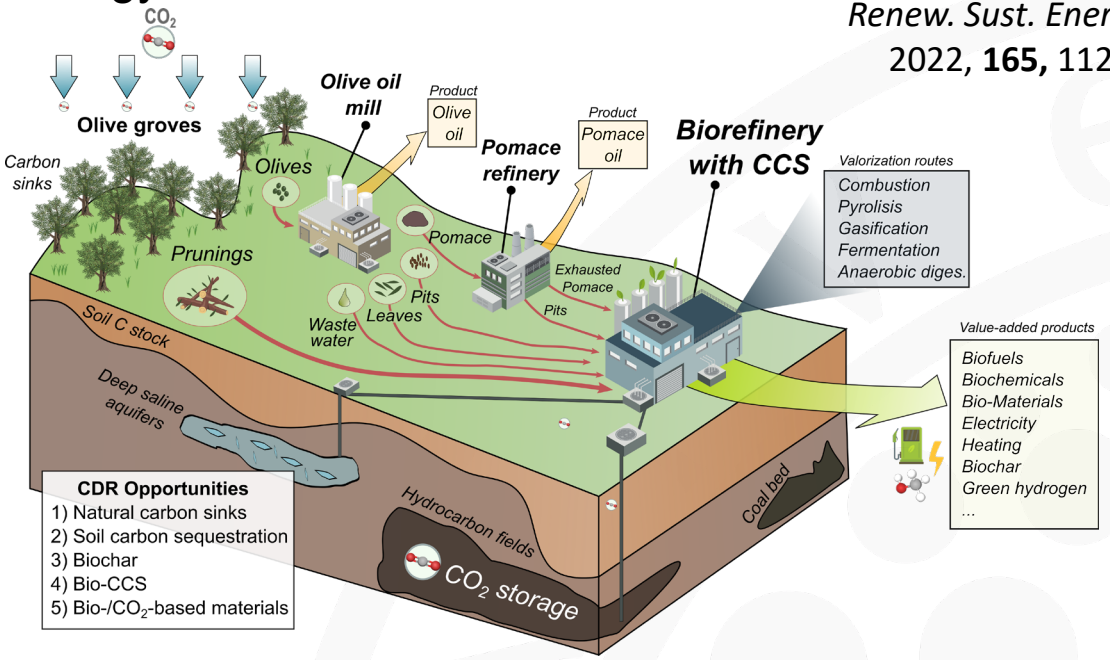


ACS Sustainable Chem. Eng. 2025, **13**, 7, 2732–2745

“Green” fuels

- **Olive oil sector** in Spain generates large amounts of **olive-derived byproducts and biomass waste** that represent valuable resources for extracting high-value compounds and clean energy.



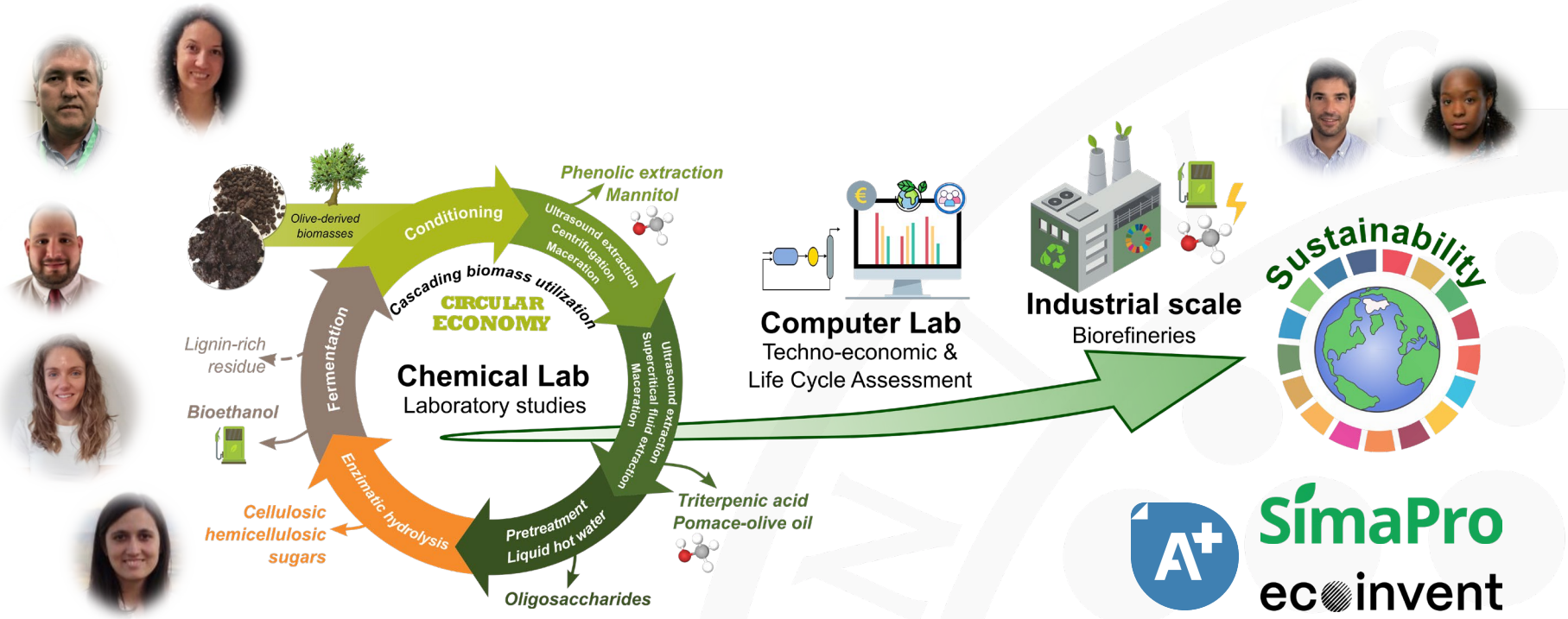


Renew. Sust. Energ. Rev.. 2022, **165,** 112609.

IDEA: To evaluate how **multiproduct biorefineries with/without CCS** can **valorize exhausted olive pomace into bio-based products** while reducing the carbon footprint and avoiding broader environmental trade-offs.

“Green” fuels

- **Two levels of research**: **experimental research** and industrial-scale **modeling/simulation exercises to analyze the technical feasibility and environmental implications** beyond climate change.

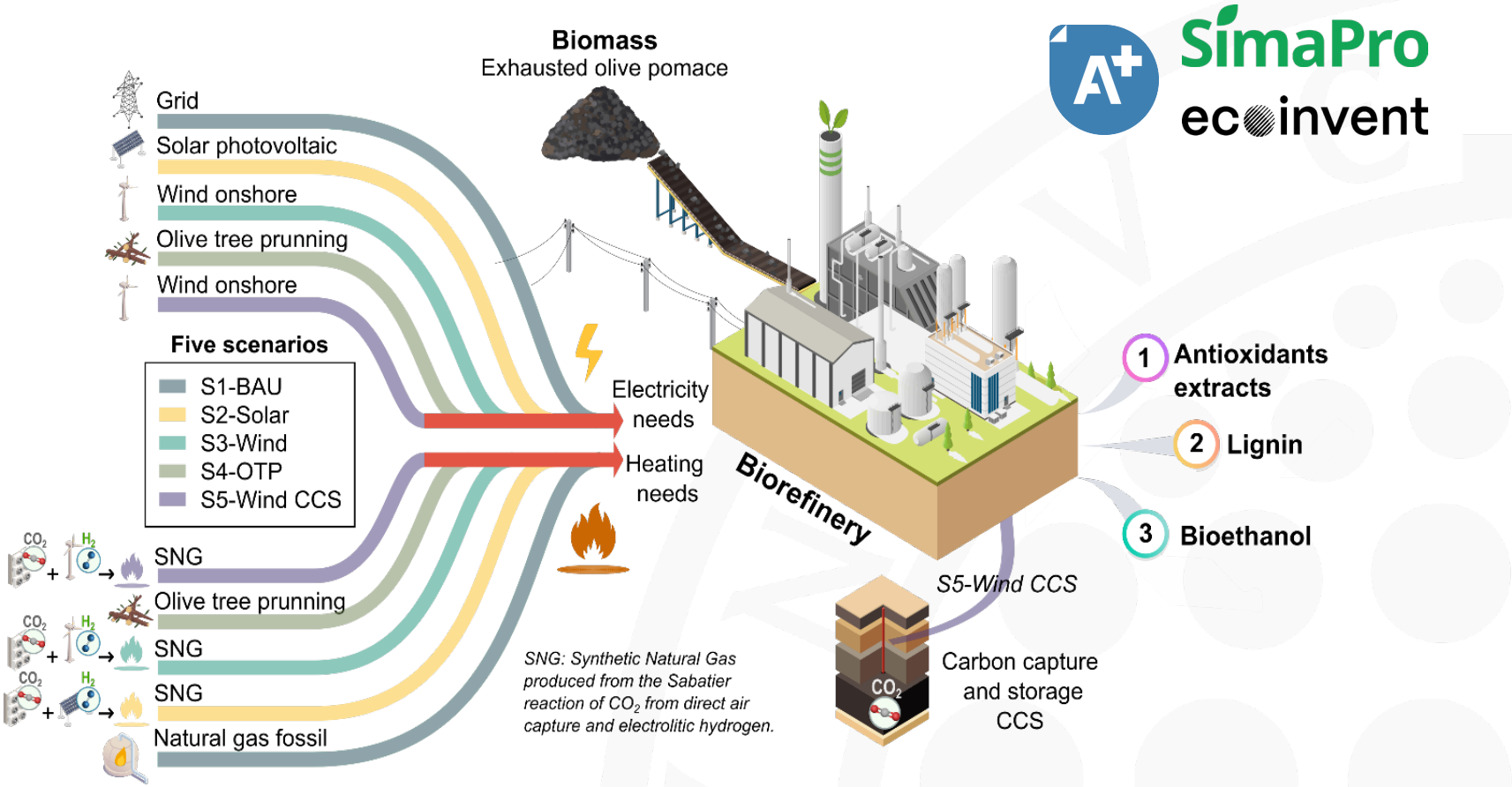


SUSPROLIBE *- SUStainable PROcesses to valorize OLIve agroindustrial biomass for moving towards a circular Bio-Economy (TED2021-132614A-I00)*

“Green” fuels

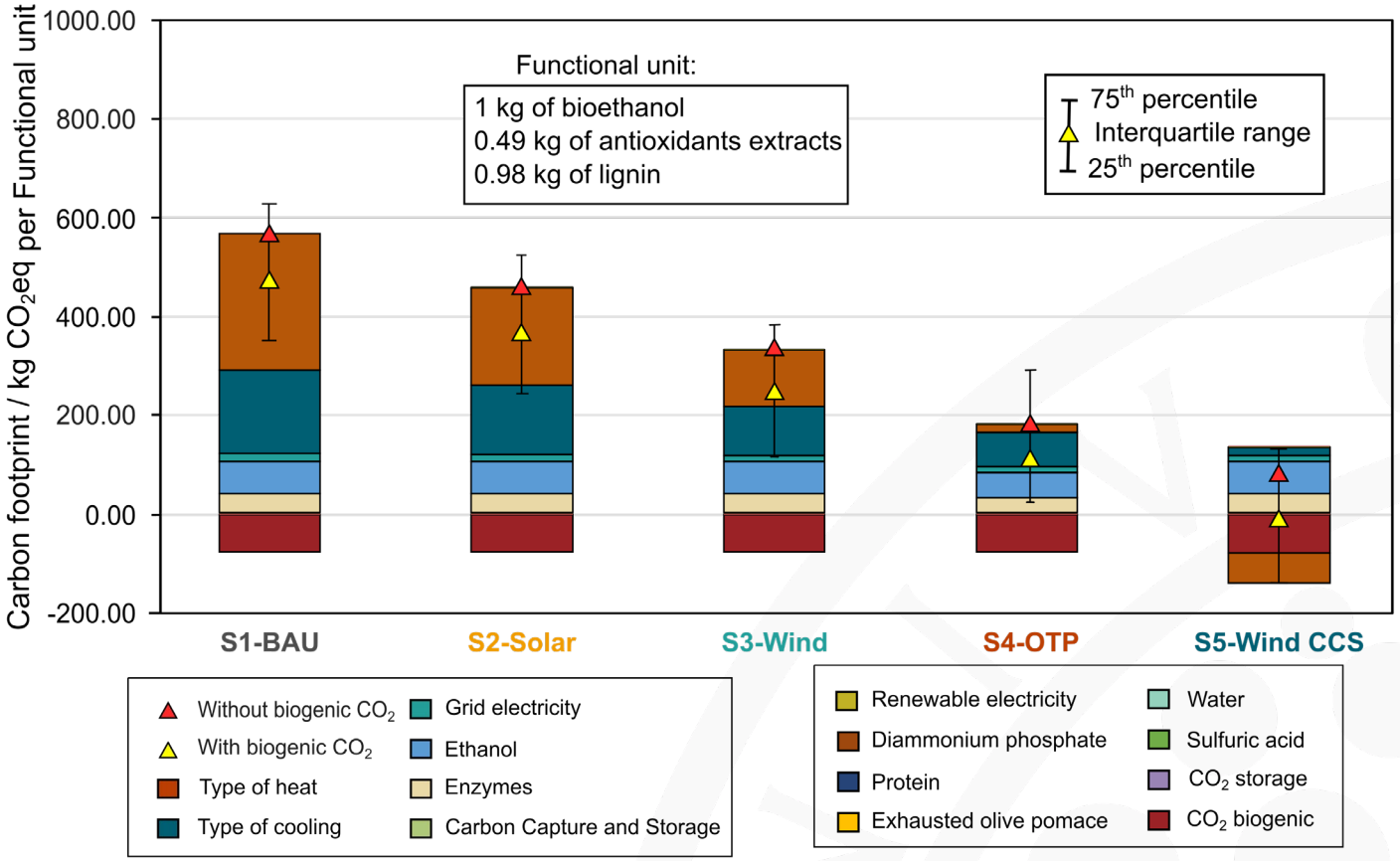
2

Evaluate five **scenarios** of the biorefinery system producing **antioxidant** extracts, **lignin**, and **bioethanol** from exhausted olive pomace (**varying electricity and heating** sources powering the system and **integrating CCS**).



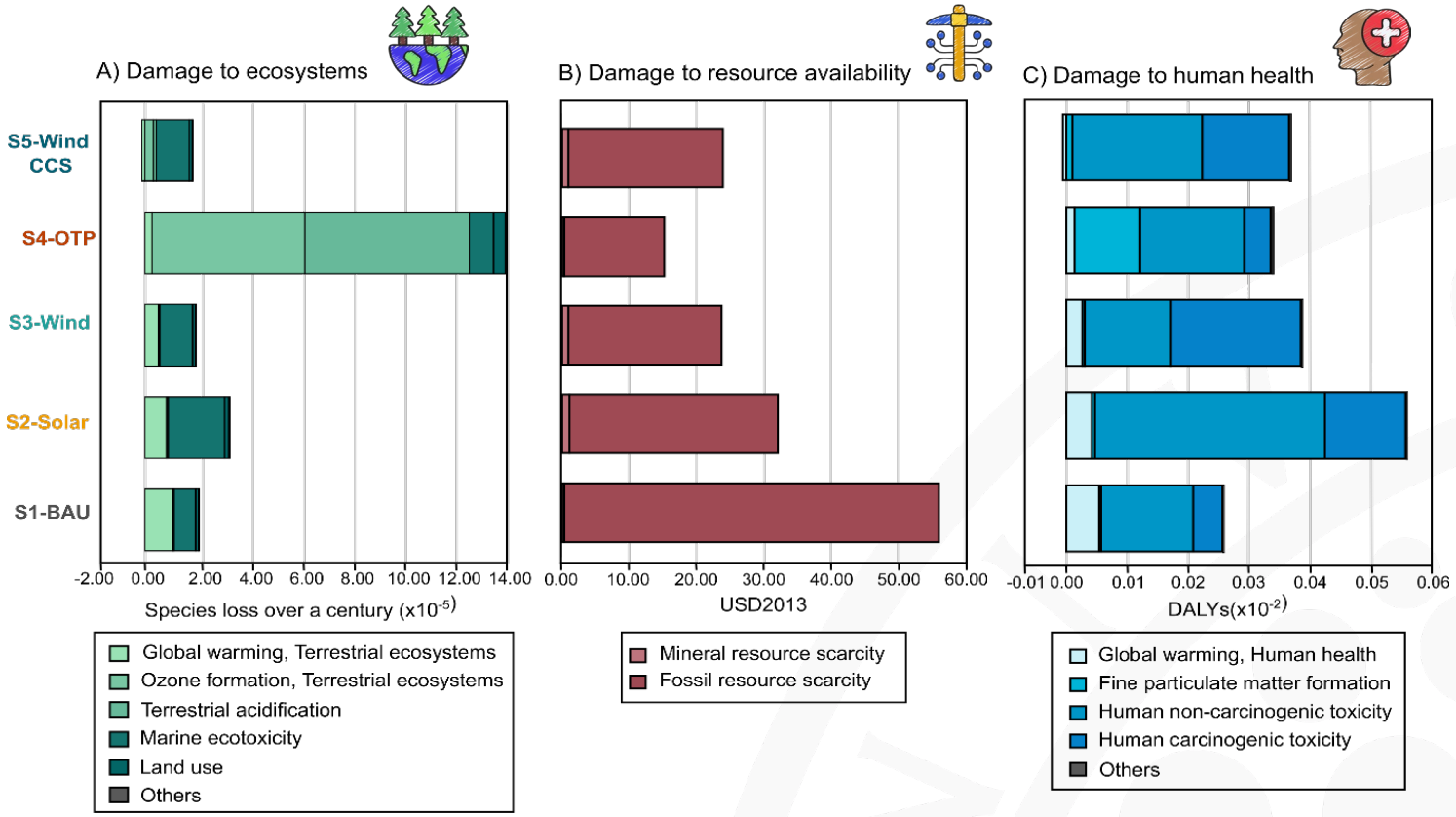
Carbon footprint

GWP100 (ReCiPe 2016)



- ✓ **Heating and cooling** demands are identified as the **main contributors**.
- ✓ **S1-BAU** shows the highest carbon footprint.
- ✓ **S5-Wind CCS** achieves a **negative carbon footprint** (*cradle-to-gate*).
- ✓ **Selling carbon credits** offer a promising strategy to offset extra costs.

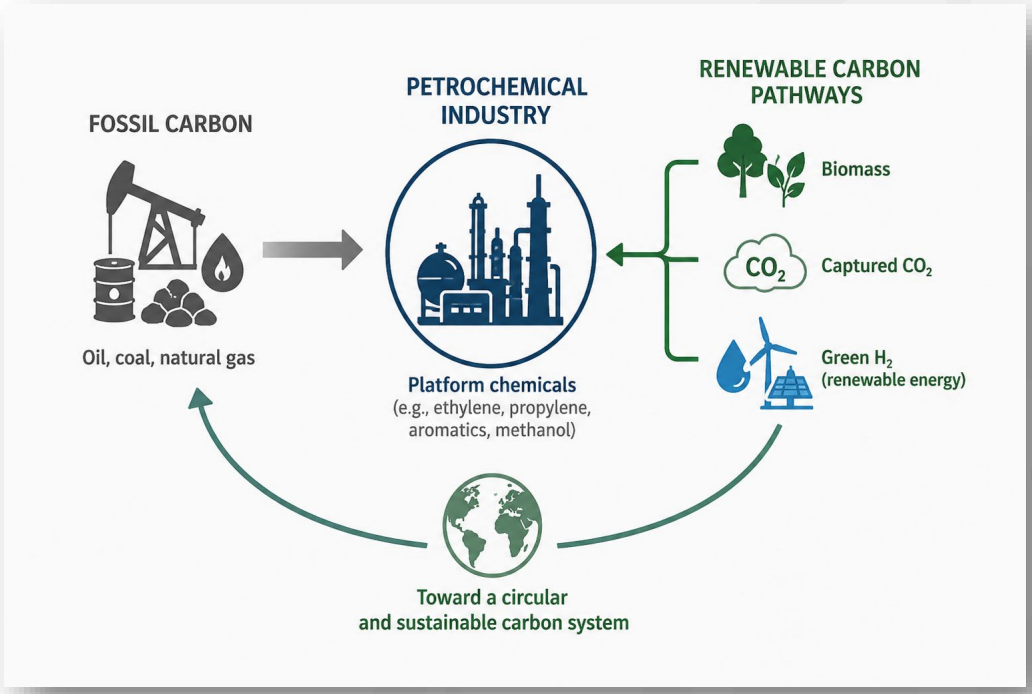
Other environmental implications



✓ Our results reveals that there may be **trade-offs to other environmental categories**, causing increased impact on **ecosystem health**, **resource availability**, and **human health**.

3 Carbon valorization

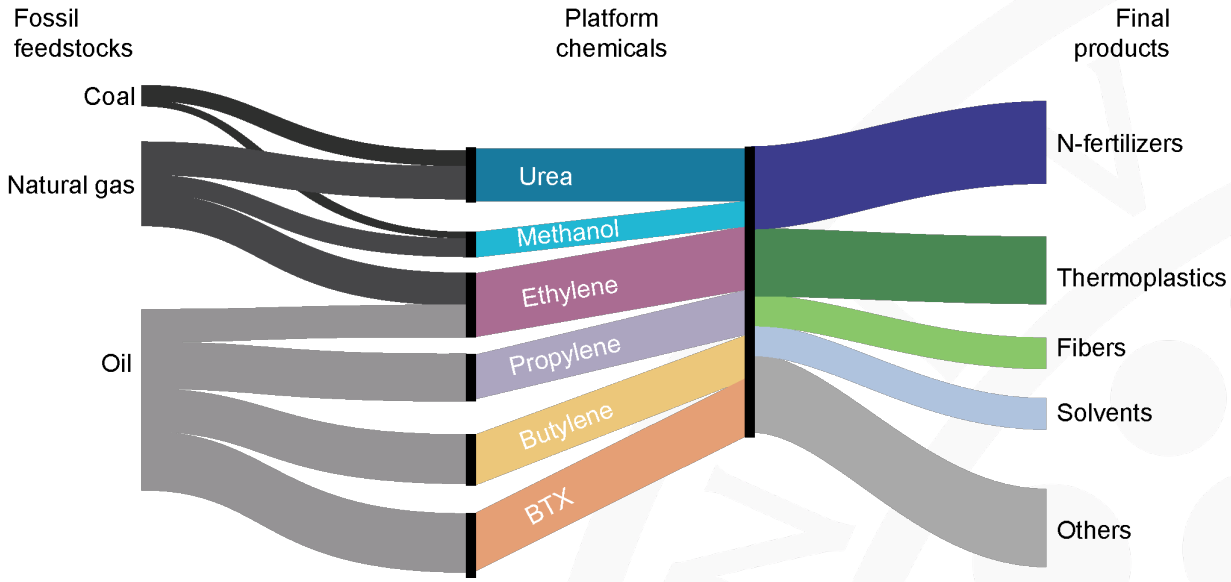
Sustainability footprints of a renewable carbon transition for the petrochemical sector within planetary boundaries



One Earth, 2021, **4**, 565–583.

Carbon Valorization

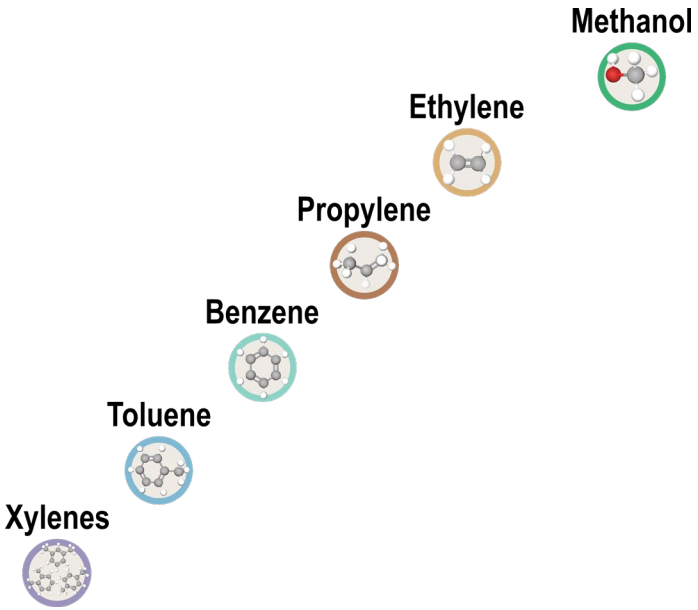
- The **chemical industry** is a **cornerstone in the economy** delivering chemicals for key industrial sectors (transportation, agriculture/food, end users…)
- **Today platform chemicals are produced from fossil resources** which makes it unsustainable.



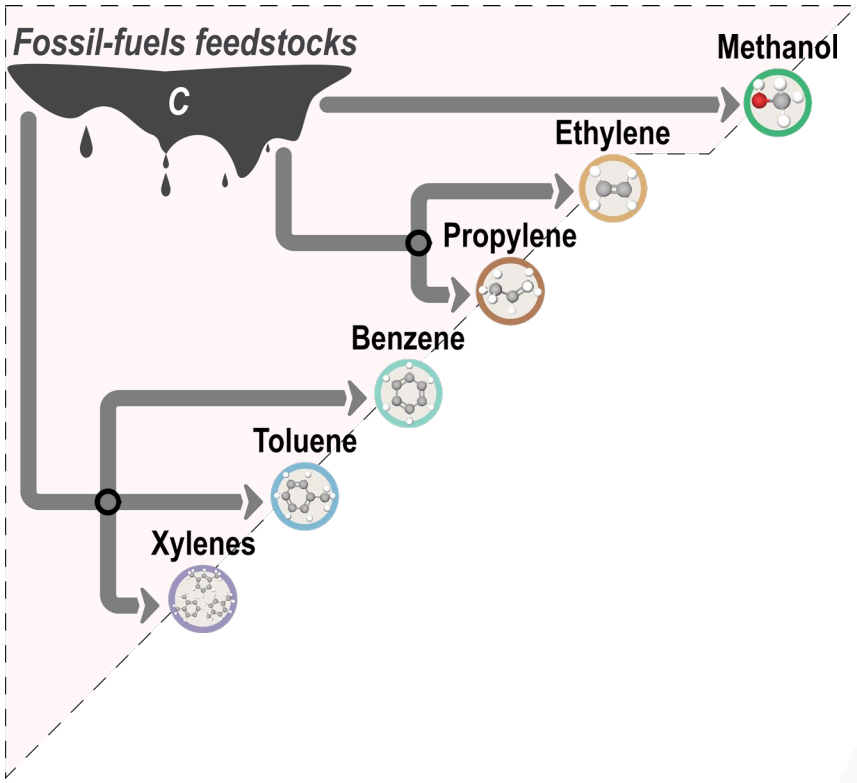
IDEA: To assess through the **lens of planetary boundaries** the transition of the **fossil-based** petrochemical industry to an analogous one based on **renewable carbon**.

Carbon Valorization

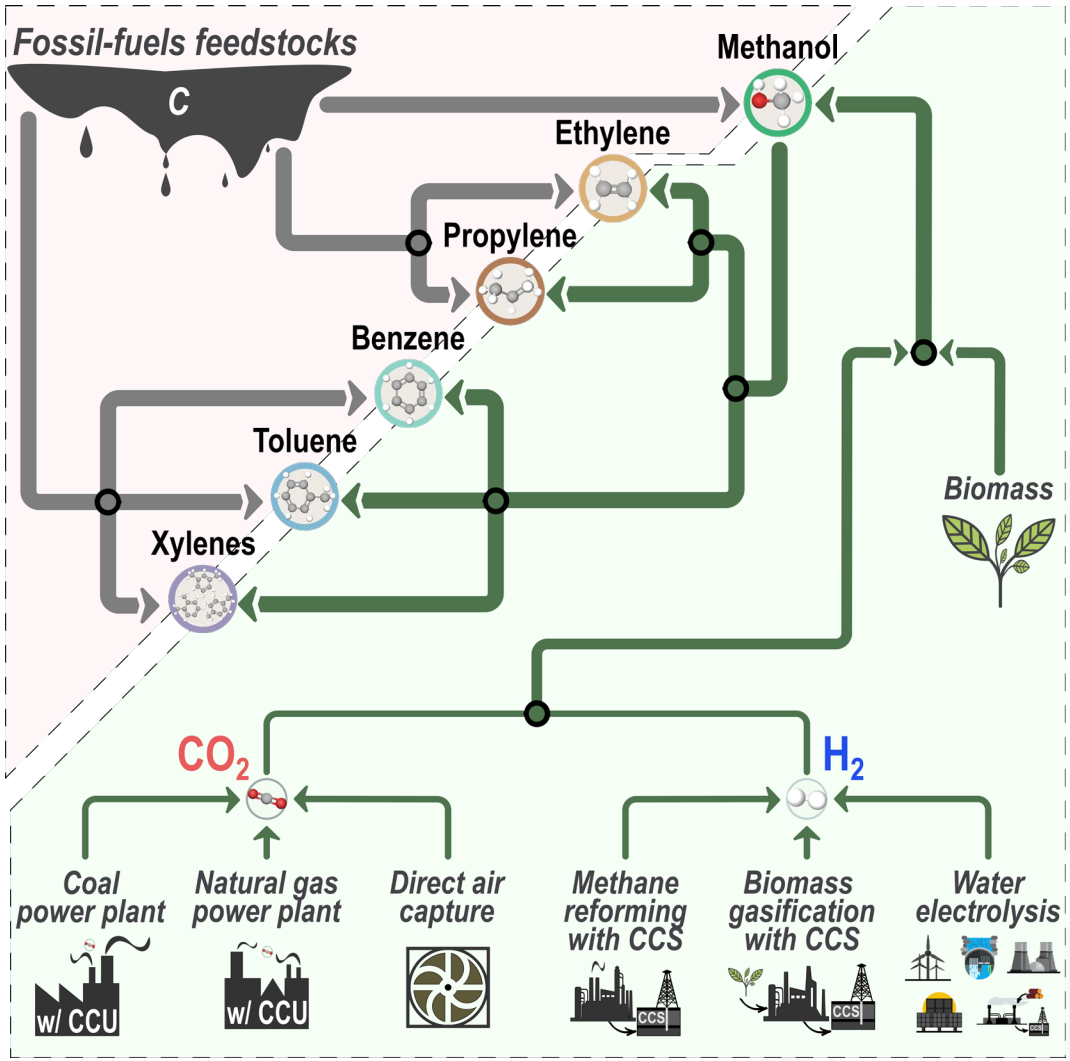
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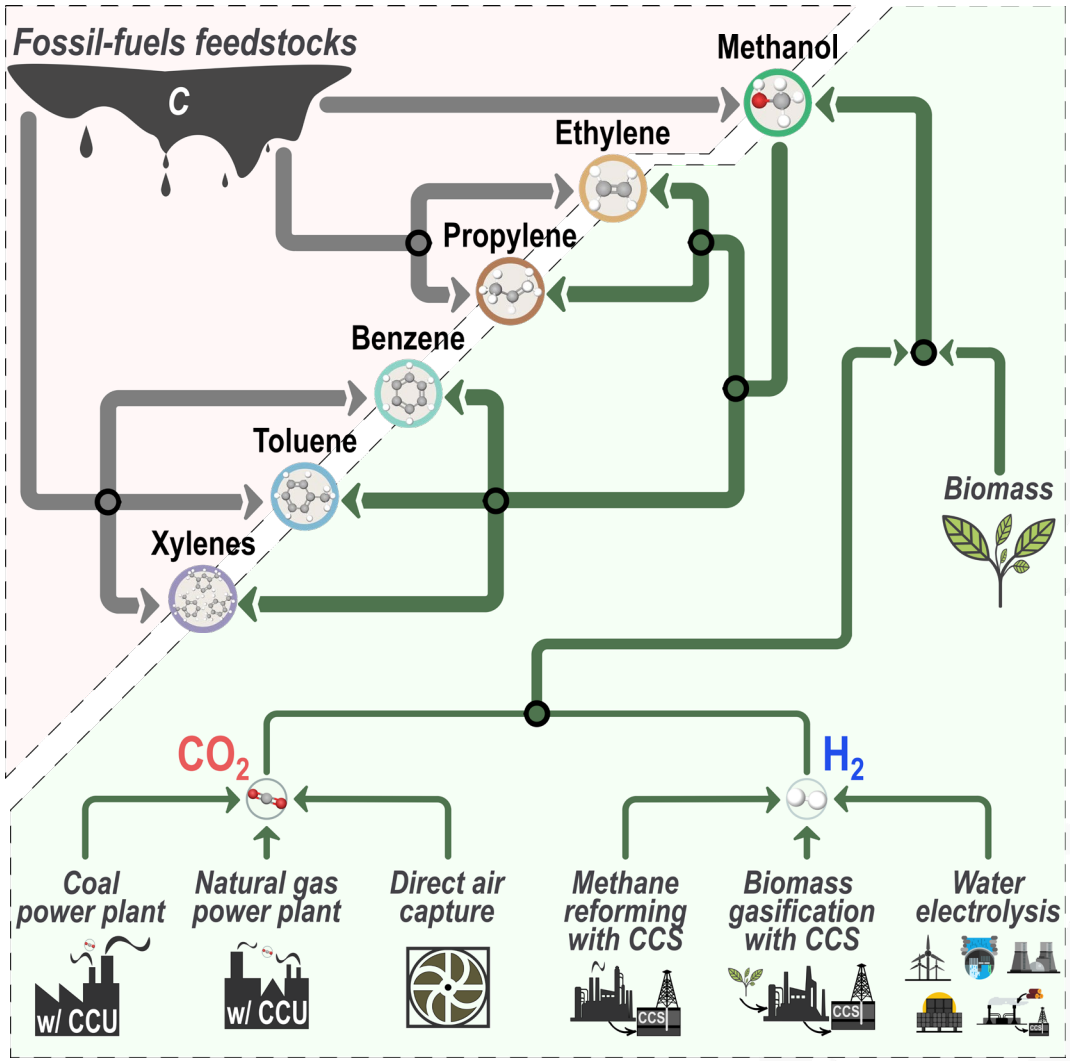
Carbon Valorization



Carbon Valorization



Carbon Valorization



Our work assesses

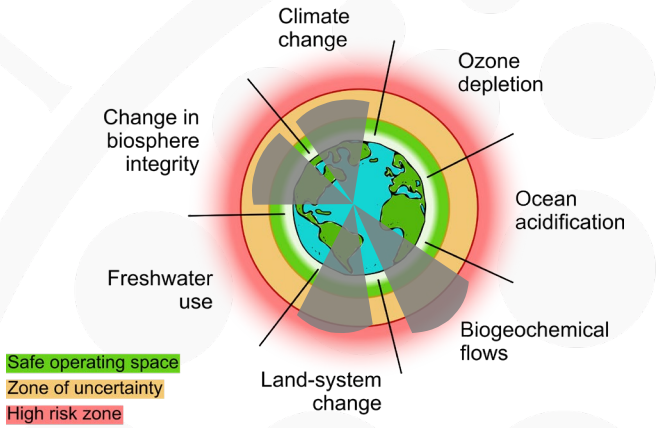
51 scenarios

50 scenarios based on

renewable carbon

vs

petrochemical industry



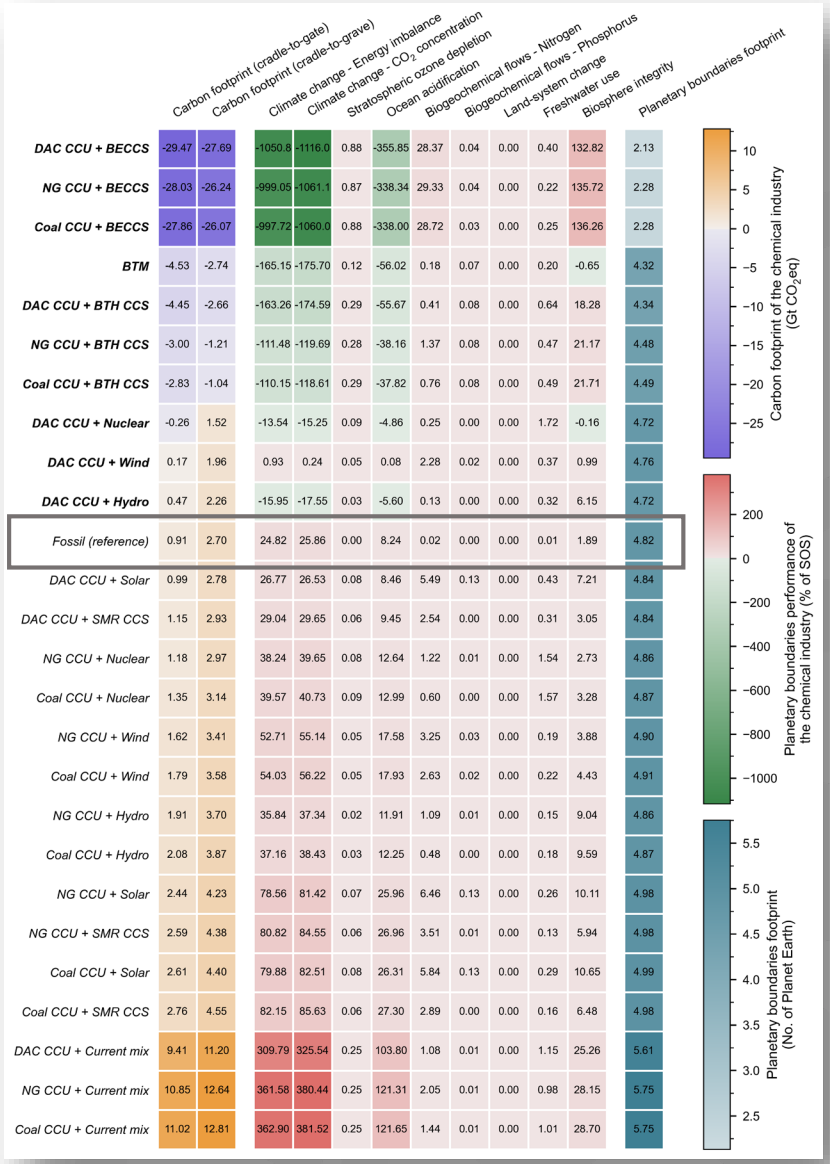
Implications of restructuring chemical industry remain **unclear**

Impact of renewable-based scenarios

Improve footprint in climate change

Current petrochemical industry

Worsen footprint in climate change

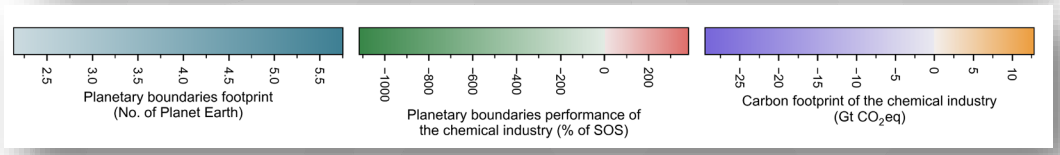


DAC CCU + BECCS	-29.47	-27.69	-1050.8	-1116.0	0.88	-355.85	28.37	0.04	0.00	0.40	132.82	2.13
NG CCU + BECCS	-28.03	-26.24	-999.05	-1061.1	0.87	-338.34	29.33	0.04	0.00	0.22	135.72	2.28
Coal CCU + BECCS	-27.86	-26.07	-997.72	-1060.0	0.88	-338.00	28.72	0.03	0.00	0.25	136.26	2.28
BTM	-4.53	-2.74	-165.15	-175.70	0.12	-56.02	0.18	0.07	0.00	0.20	-0.65	4.32
DAC CCU + BTH CCS	-4.45	-2.66	-163.26	-174.59	0.29	-55.67	0.41	0.08	0.00	0.64	18.28	4.34
NG CCU + BTH CCS	-3.00	-1.21	-111.48	-119.69	0.28	-38.16	1.37	0.08	0.00	0.47	21.17	4.48
Coal CCU + BTH CCS	-2.83	-1.04	-110.15	-118.61	0.29	-37.82	0.76	0.08	0.00	0.49	21.71	4.49
DAC CCU + Nuclear	-0.26	1.52	-13.54	-15.25	0.09	-4.86	0.25	0.00	0.00	1.72	-0.16	4.72
DAC CCU + Wind	0.17	1.96	0.93	0.24	0.05	0.08	2.28	0.02	0.00	0.37	0.99	4.76
DAC CCU + Hydro	0.47	2.26	-15.95	-17.55	0.03	-5.60	0.13	0.00	0.00	0.32	6.15	4.72
Fossil (reference)	0.91	2.70	24.82	25.86	0.00	8.24	0.02	0.00	0.00	0.01	1.89	4.82
DAC CCU + Solar	0.99	2.78	26.77	26.53	0.08	8.46	5.49	0.13	0.00	0.43	7.21	4.84
DAC CCU + SMR CCS	1.15	2.93	29.04	29.65	0.06	9.45	2.54	0.00	0.00	0.31	3.05	4.84
NG CCU + Nuclear	1.18	2.97	38.24	39.65	0.08	12.64	1.22	0.01	0.00	1.54	2.73	4.86
Coal CCU + Nuclear	1.35	3.14	39.57	40.73	0.09	12.99	0.60	0.00	0.00	1.57	3.28	4.87
NG CCU + Wind	1.62	3.41	52.71	55.14	0.05	17.58	3.25	0.03	0.00	0.19	3.88	4.90
Coal CCU + Wind	1.79	3.58	54.03	56.22	0.05	17.93	2.63	0.02	0.00	0.22	4.43	4.91
NG CCU + Hydro	1.91	3.70	35.84	37.34	0.02	11.91	1.09	0.01	0.00	0.15	9.04	4.86
Coal CCU + Hydro	2.08	3.87	37.16	38.43	0.03	12.25	0.48	0.00	0.00	0.18	9.59	4.87
NG CCU + Solar	2.44	4.23	78.56	81.42	0.07	25.96	6.46	0.13	0.00	0.26	10.11	4.98
NG CCU + SMR CCS	2.59	4.38	80.82	84.55	0.06	26.96	3.51	0.01	0.00	0.13	5.94	4.98
Coal CCU + Solar	2.61	4.40	79.88	82.51	0.08	26.31	5.84	0.13	0.00	0.29	10.65	4.99
Coal CCU + SMR CCS	2.76	4.55	82.15	85.63	0.06	27.30	2.89	0.00	0.00	0.16	6.48	4.98
DAC CCU + Current mix	9.41	11.20	309.79	325.54	0.25	103.80	1.08	0.01	0.00	1.15	25.26	5.61
NG CCU + Current mix	10.85	12.64	361.58	380.44	0.25	121.31	2.05	0.01	0.00	0.98	28.15	5.75
Coal CCU + Current mix	11.02	12.81	362.90	381.52	0.25	121.65	1.44	0.01	0.00	1.01	28.70	5.75

Impact of renewable-based scenarios

- Large CO_2 savings. Reduction of carbon footprint.
- Some routes even lead to CO_2 removal (negative emissions).
- Burden-shifting emerge.
- DAC CCU + BECCS exceeds the biodiversity PB threshold by 30%.

	Carbon footprint (cradle-to-gate)	Carbon footprint (cradle-to-grave)	Climate change - Energy imbalance	Climate change - CO_2 concentration	Stratospheric ozone depletion	Ocean acidification	Biogeochemical flows - Nitrogen	Biogeochemical flows - Phosphorus	Land-system change	Freshwater use	Biosphere integrity	Planetary
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Conclusions

Renewable, bio-based, recycling or circular solutions **do not automatically guarantee sustainability**.

Systems modelling and **Life Cycle Assessment are essential** to identify hidden trade-offs and avoid burden shifting across environmental categories.

Sustainable transitions in materials, fuels, and traditional industries require a **systems-level perspective beyond carbon footprint alone**.

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Sustainable transitions in materials, fuels, and traditional industries require a **systems-level perspective beyond carbon footprint alone**.

The challenge is no longer simply developing “green” technologies, but designing systems that are truly (environmentally) sustainable.

Thank you!

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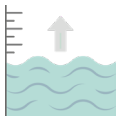
Different climate futures

Most scenarios aligned with the **Paris goal** require:

- Drastic reduction of greenhouse gas emissions.
- Carbon Dioxide Removal to compensate emissions exceeding the carbon budget and to offset emissions from hard-to-abate sectors.

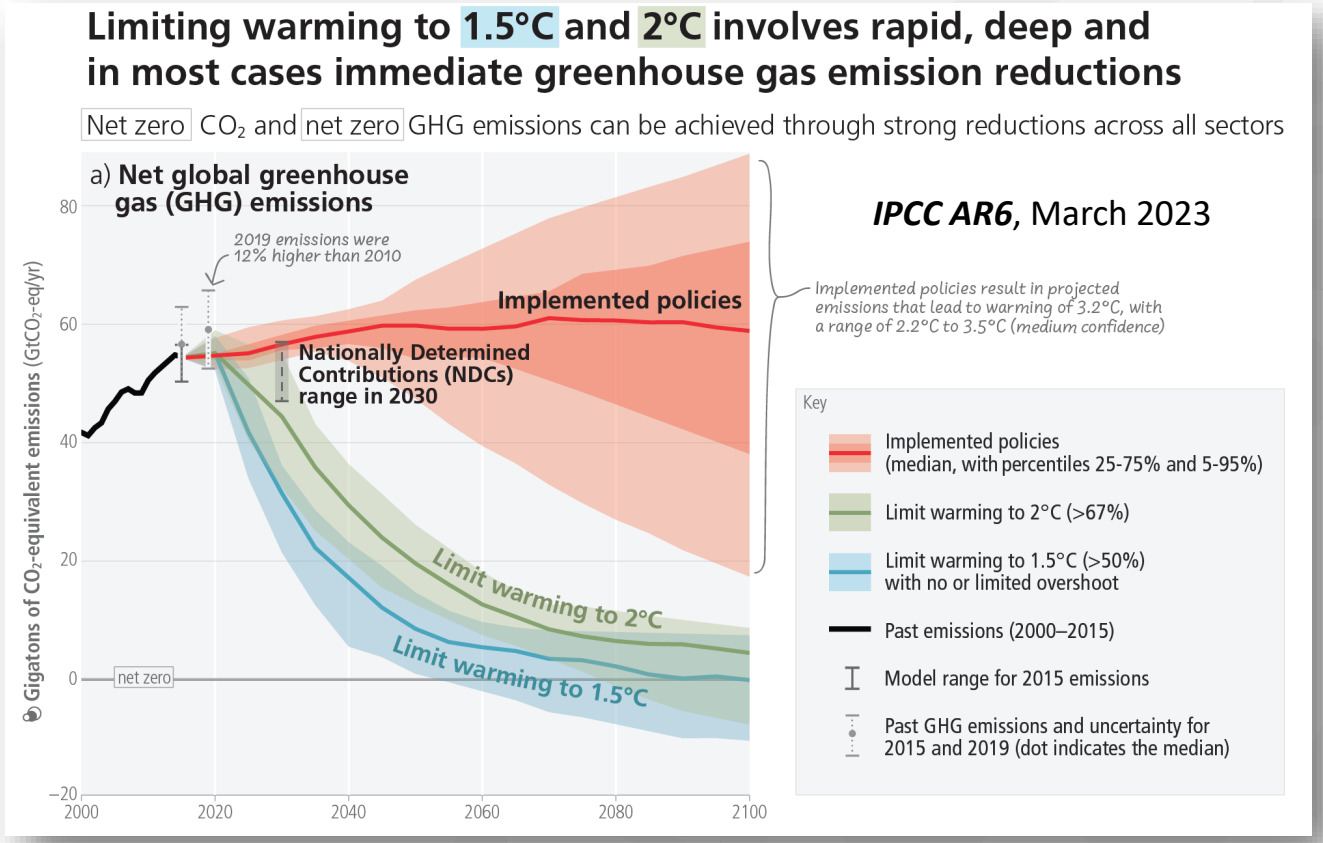












Understanding what is biochar

Biochar is a stable carbon-rich solid material primarily used for soil amendment, carbon sequestration, and environmental remediation.



Understanding what is biochar

Biochar is a stable carbon-rich solid material primarily used for soil amendment, carbon sequestration, and environmental remediation.



Forestry residues
Agricultural residues
Energy crops
Municipal solid waste
…



Gasification
Pyrolysis

1. ***Syngas (gas)***
   $CO + H_2$
2. ***Bio-oil (liquid)***
3. ***Biochar (solid)***

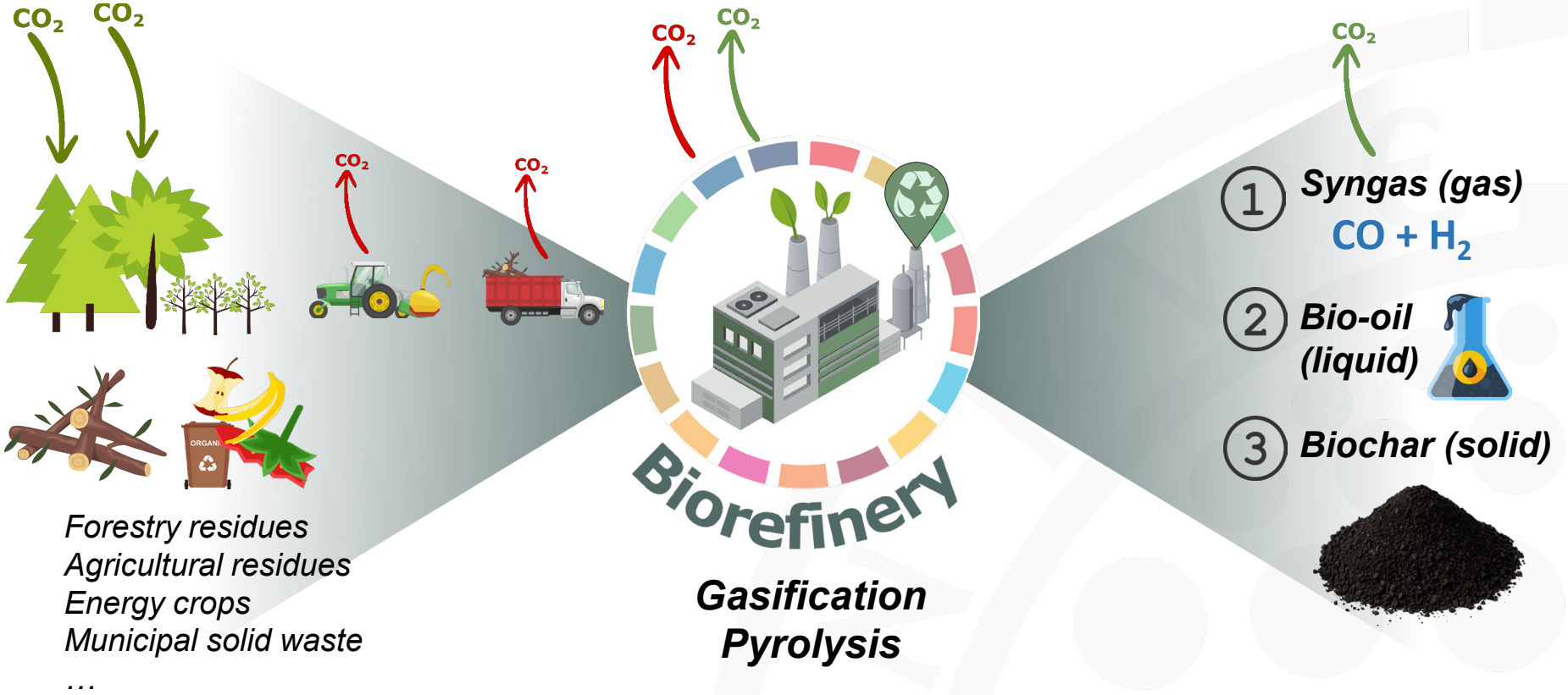


Pyrolysis: Thermal decomposition in the **absence of oxygen**, typically at 400–800°C.

Gasification: Partial oxidation at 700–1200°C in a controlled oxygen, air, or steam.

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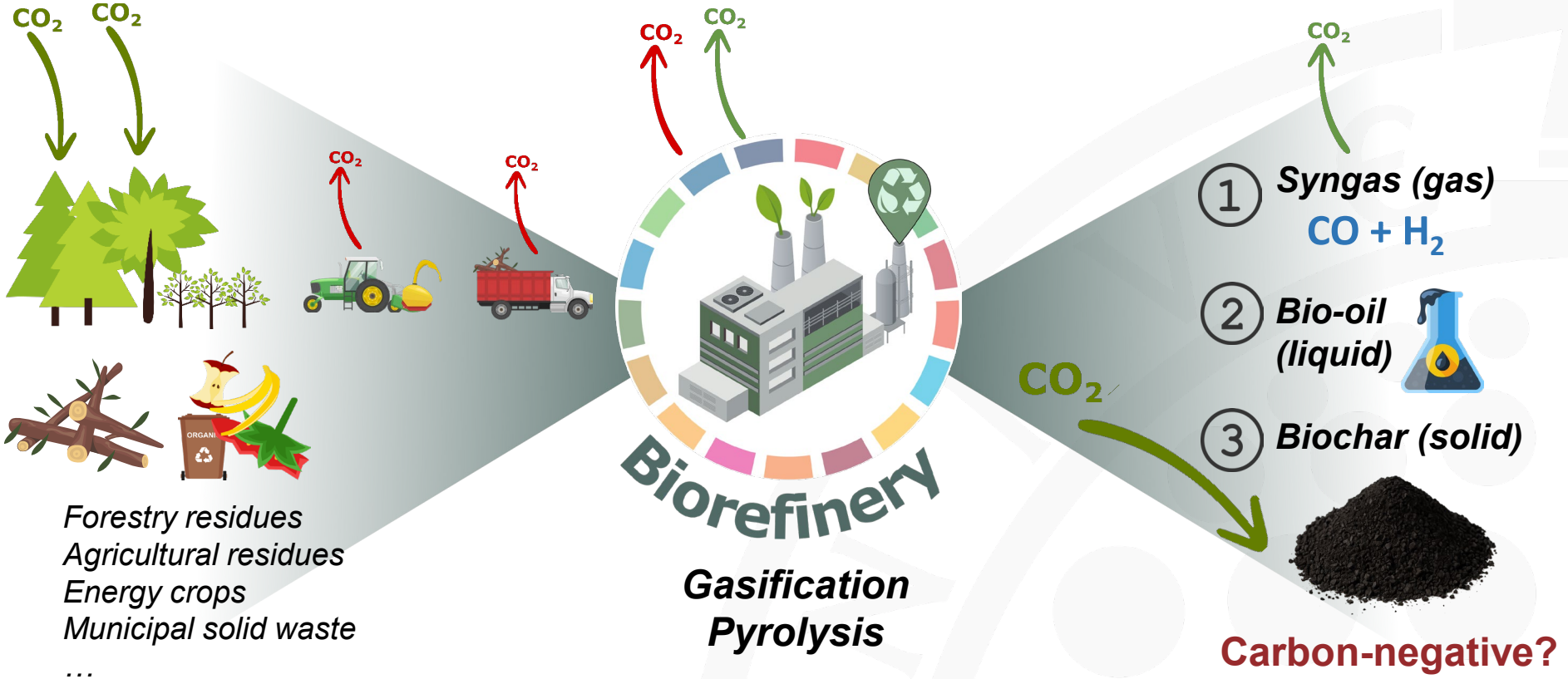


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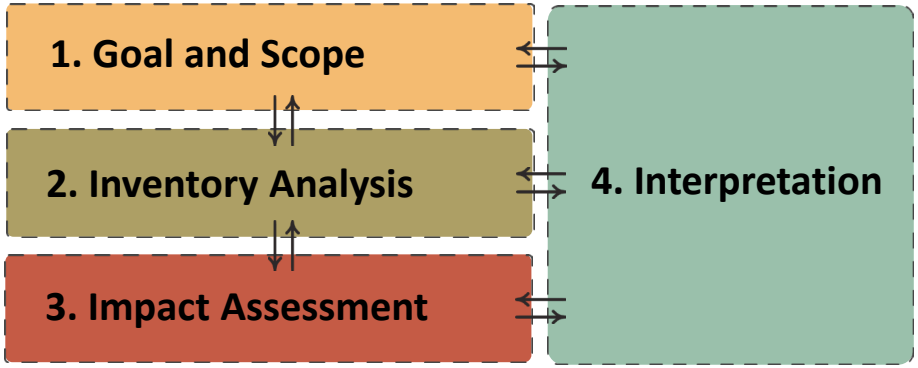


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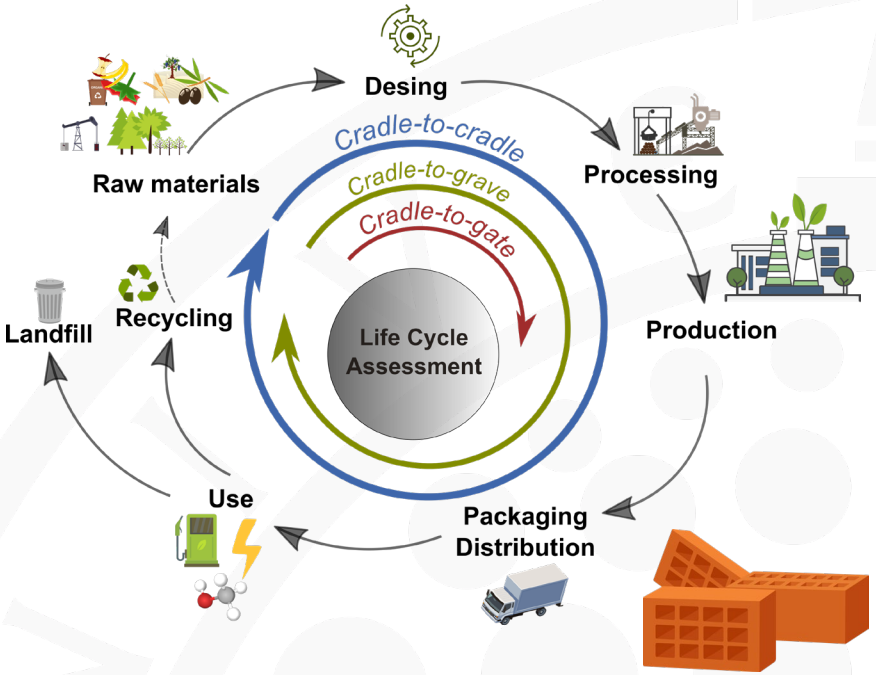
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Life Cycle Assessment (LCA)

Standarized methodology

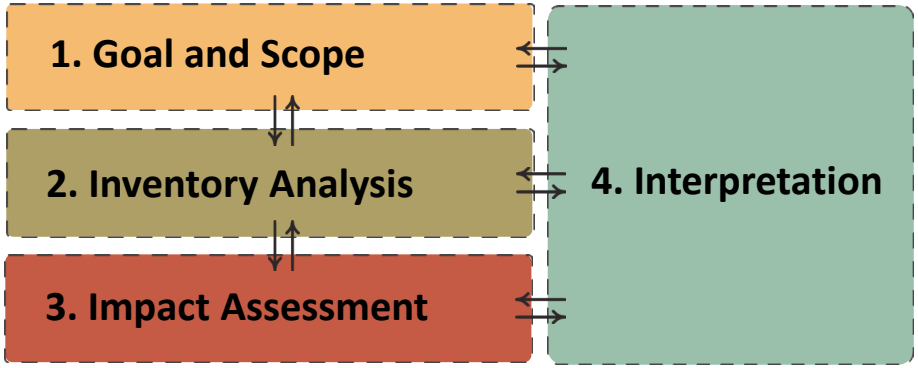


* following ISO 14040/44:2006 standards



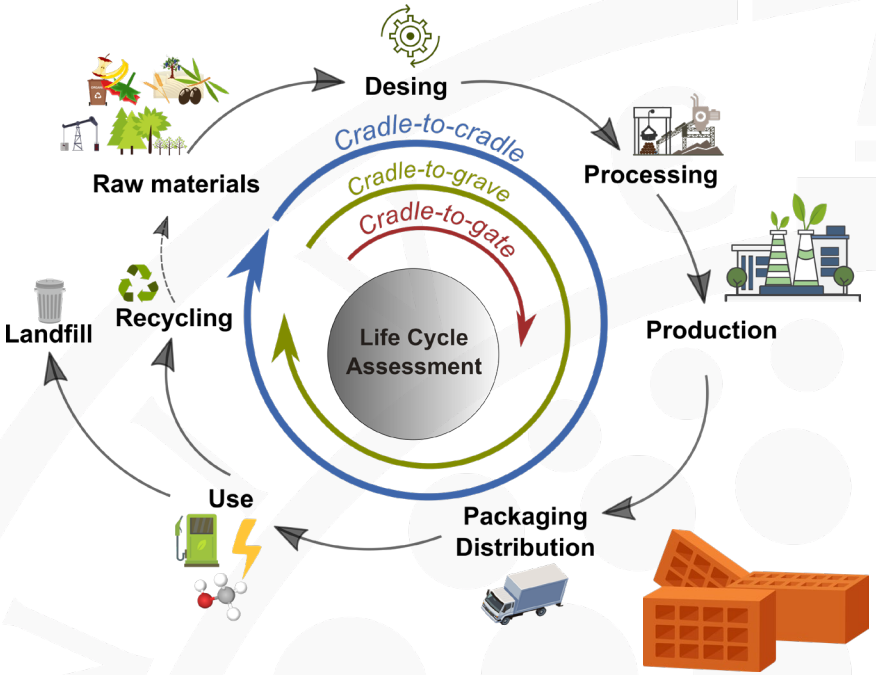
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Standarized methodology



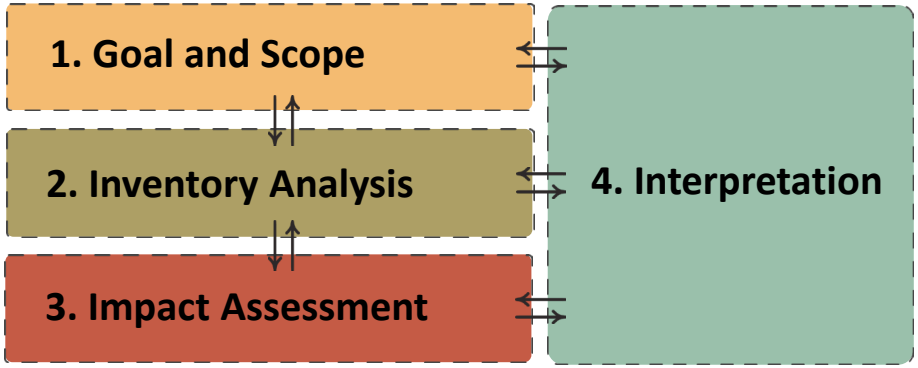
* following ISO 14040/44:2006 standards

***System boundaries**: cradle-to-gate*
***Functional unit**: 1 ton of geopolymer*



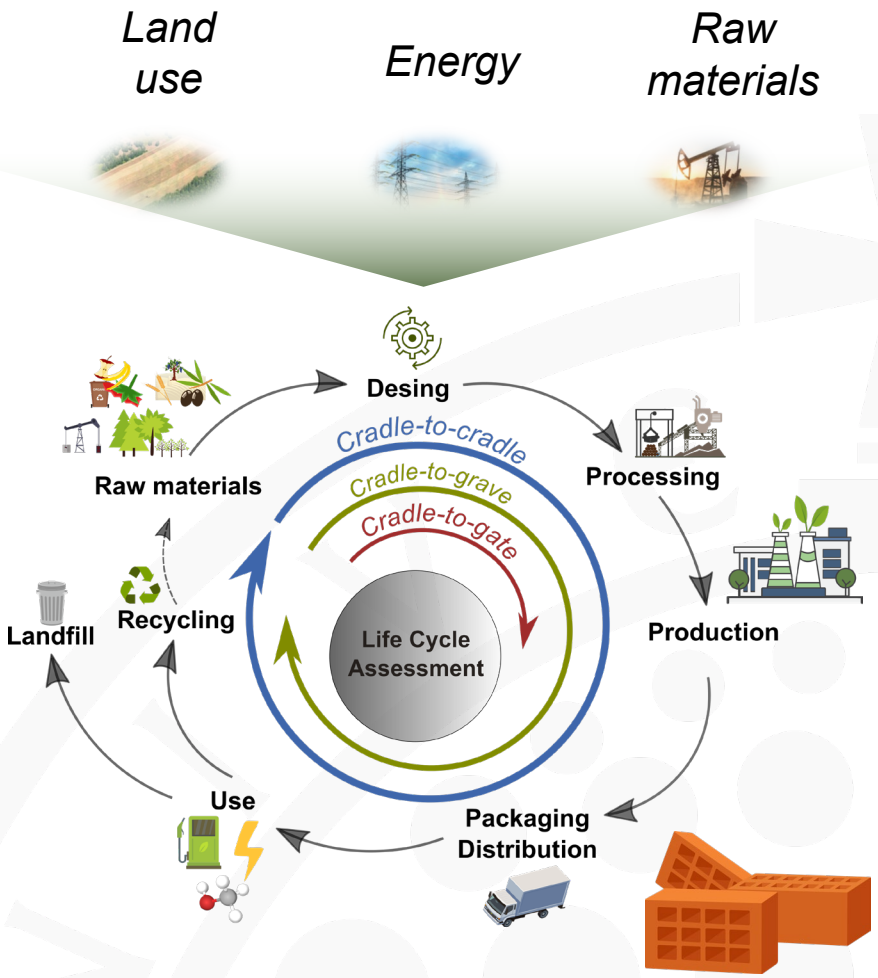
Life Cycle Assessment (LCA)

Standarized methodology



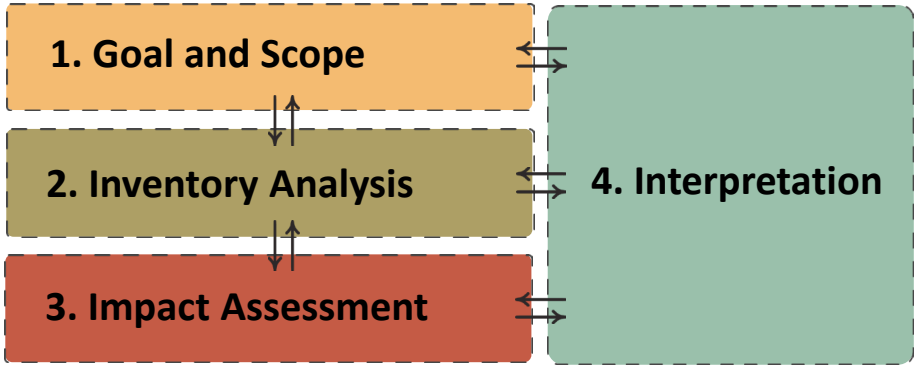
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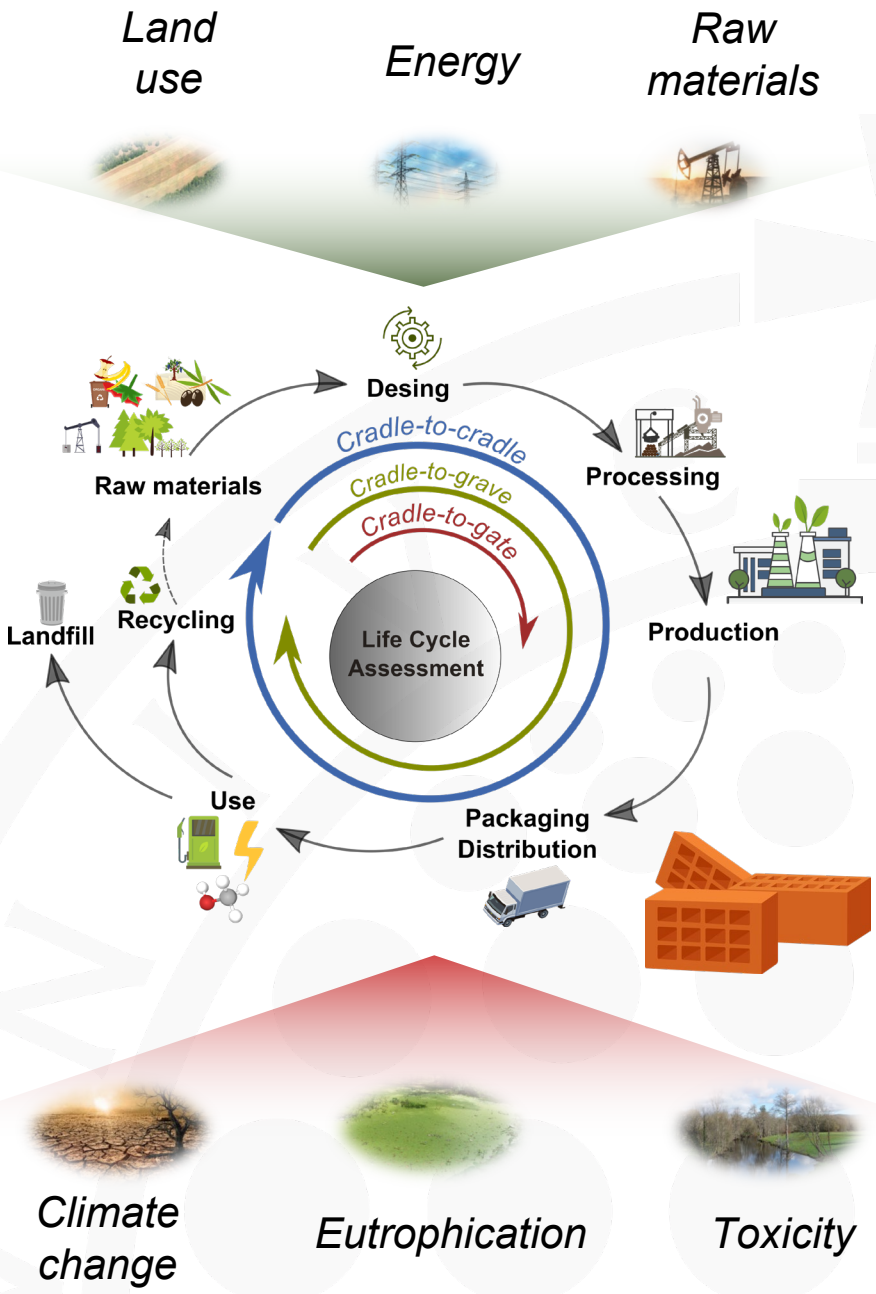
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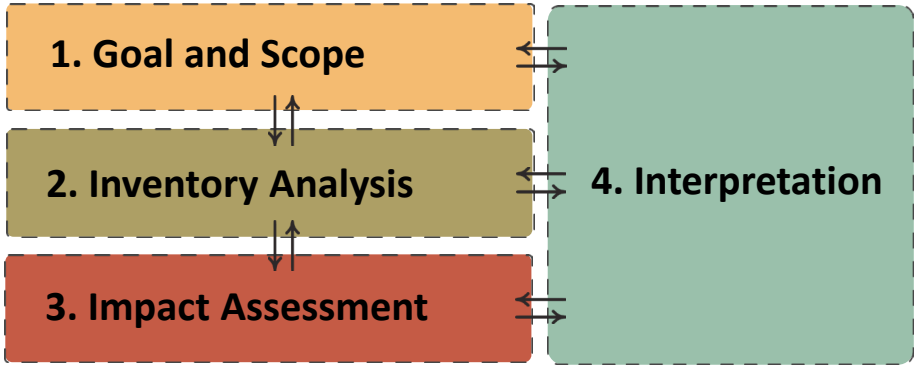
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System boundaries: *cradle-to-gate*
Functional unit: *1 ton of geopolymer*
Methodology: *ReCiPe 2016*
Indicator: *GWP100*



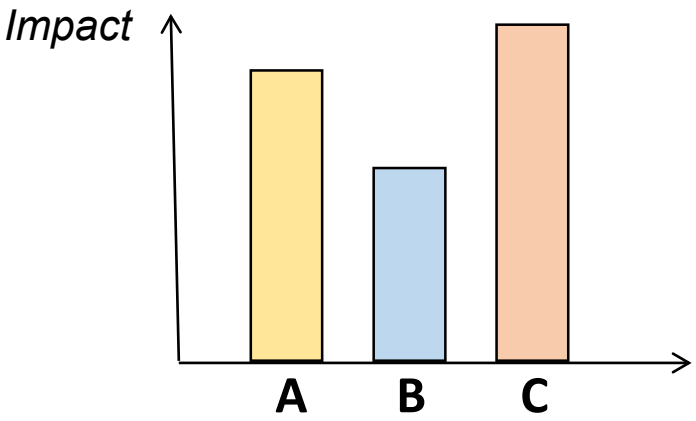
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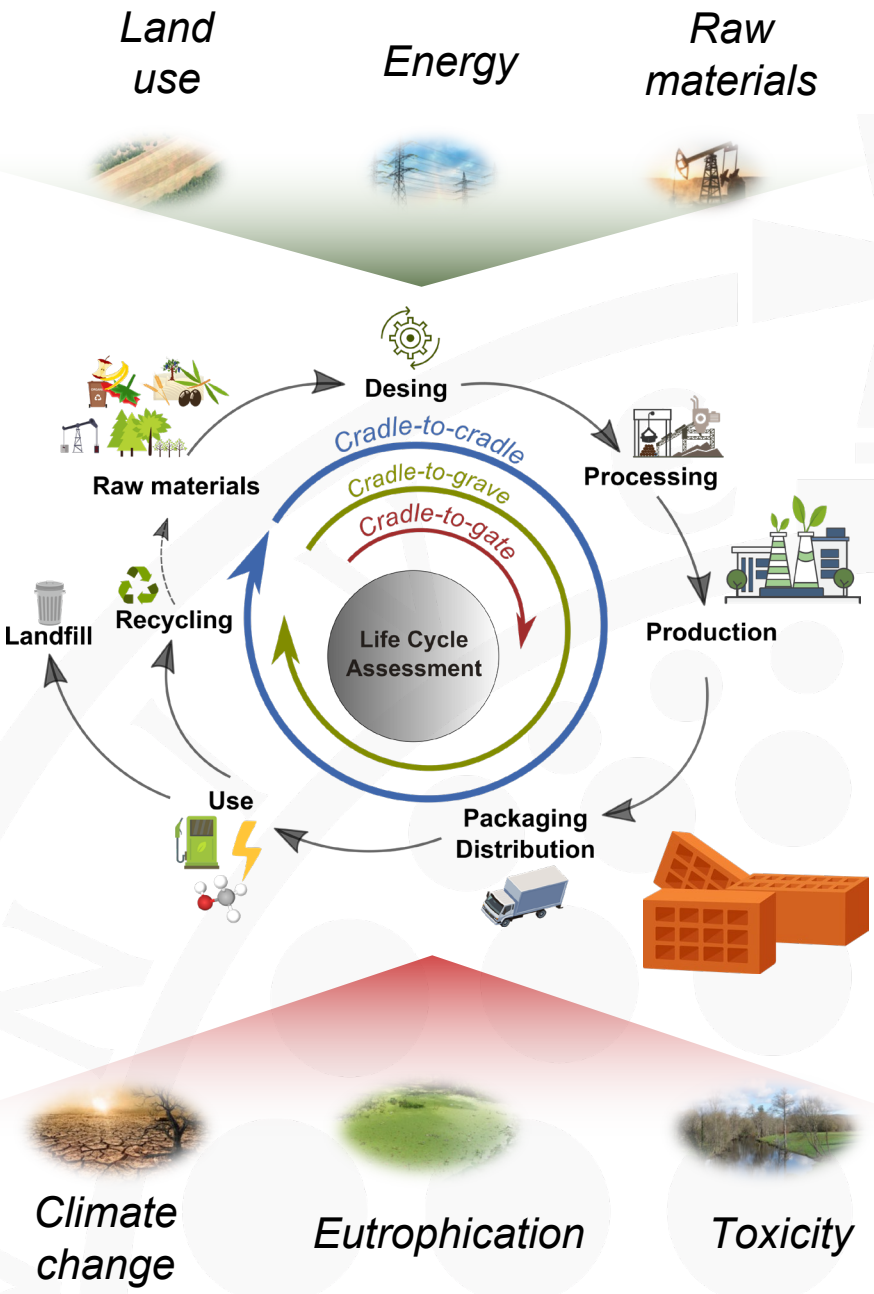
Standarized methodology



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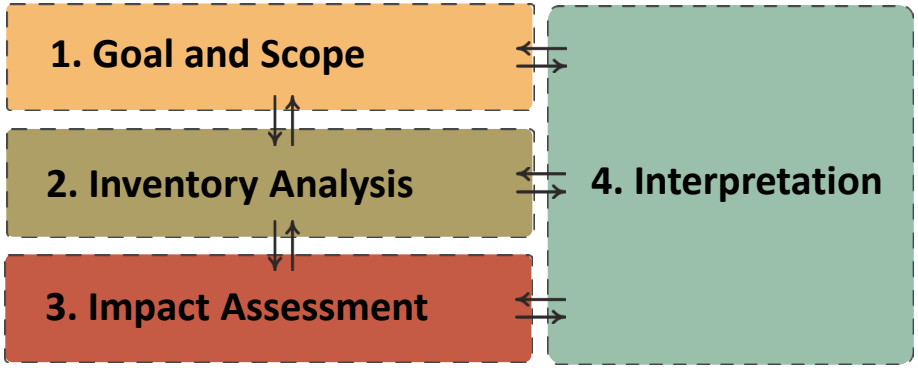
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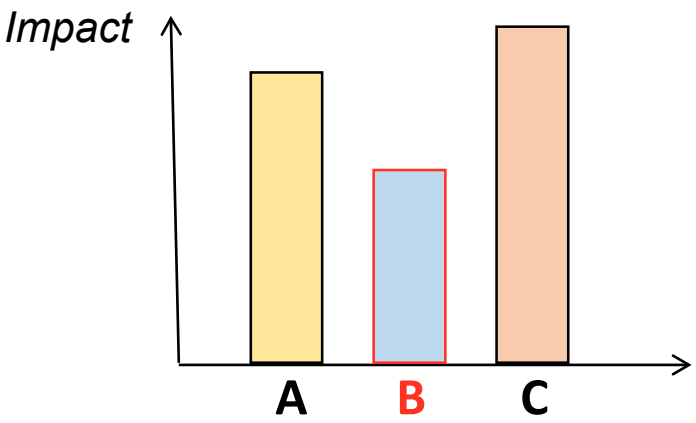
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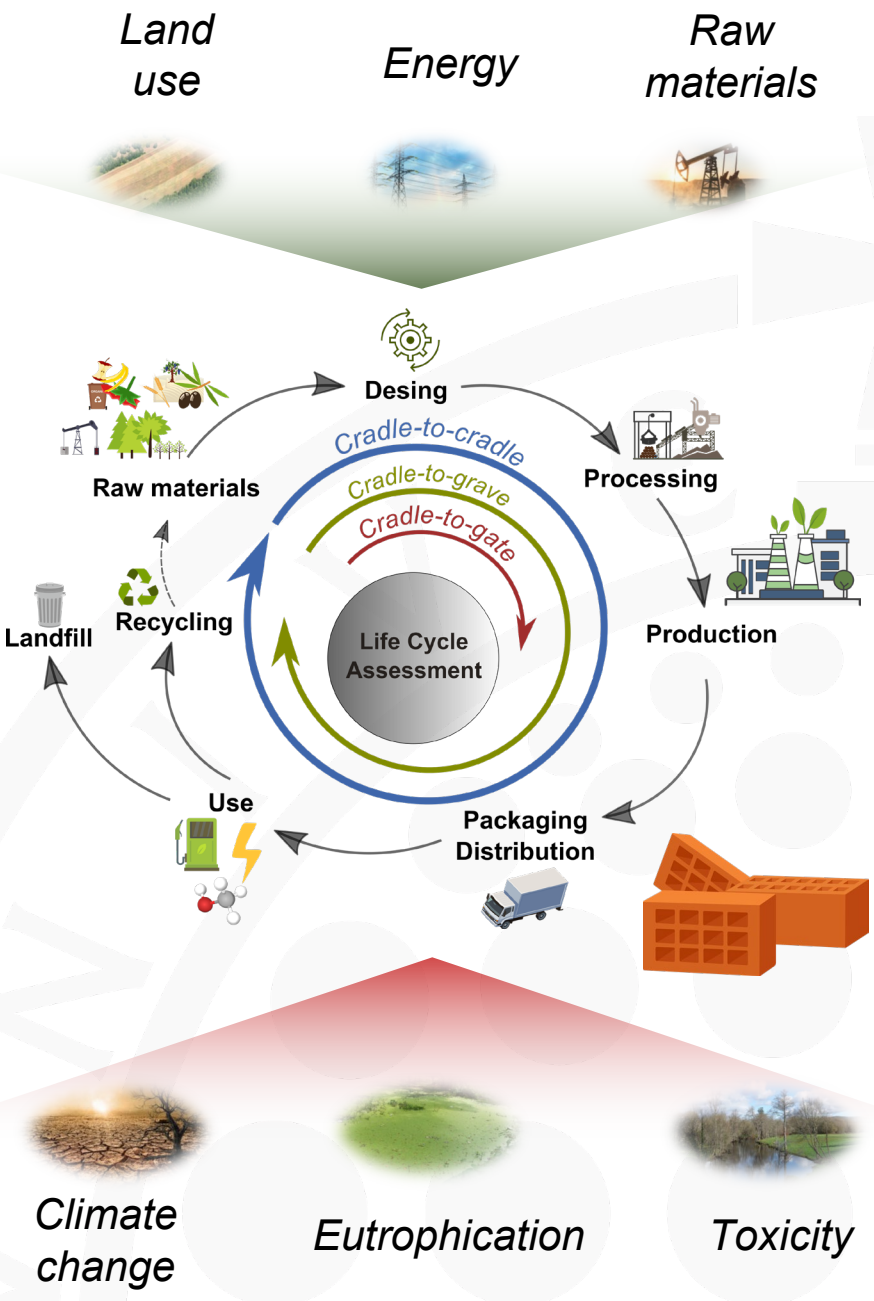
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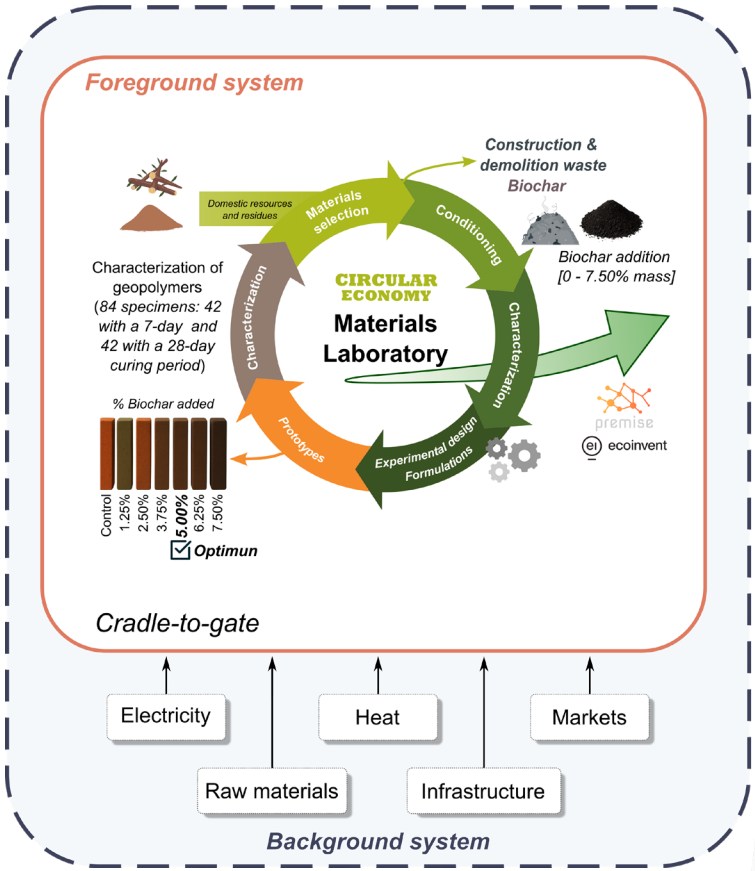
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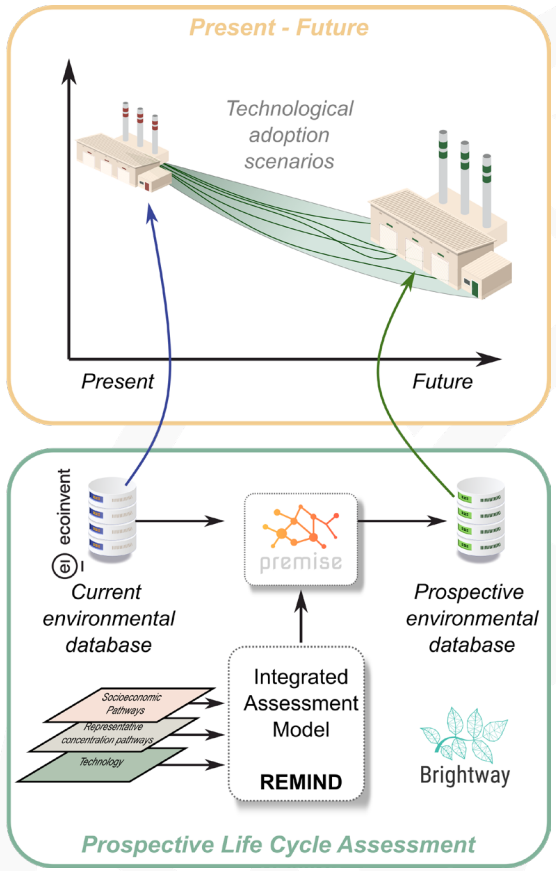




Prospective LCA

- Quantify the **long-term environmental performance** of the developed materials.
- **Functional unit** of 1 tonne of geopolymer produced and GWP 100 years.
- Future **energy projections** using the **premise** toolbox.





premise

Brightway

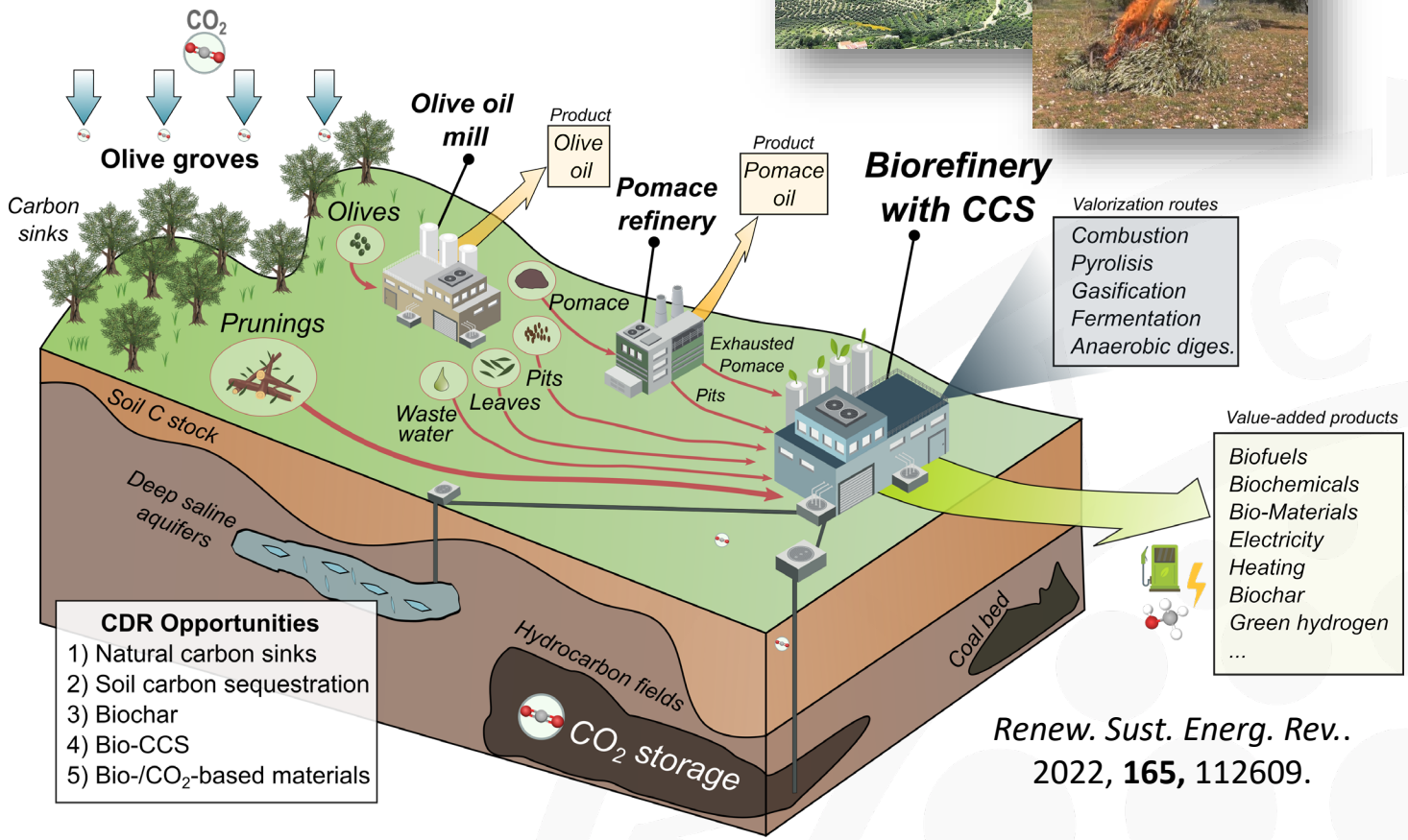
- REMIND IAM.
- SSP2–RCP1.9 scenario: socioeconomic pathway with achieving carbon neutrality by mid-century.

ecoinvent *v3.8 database*

Olive-derived biomasses





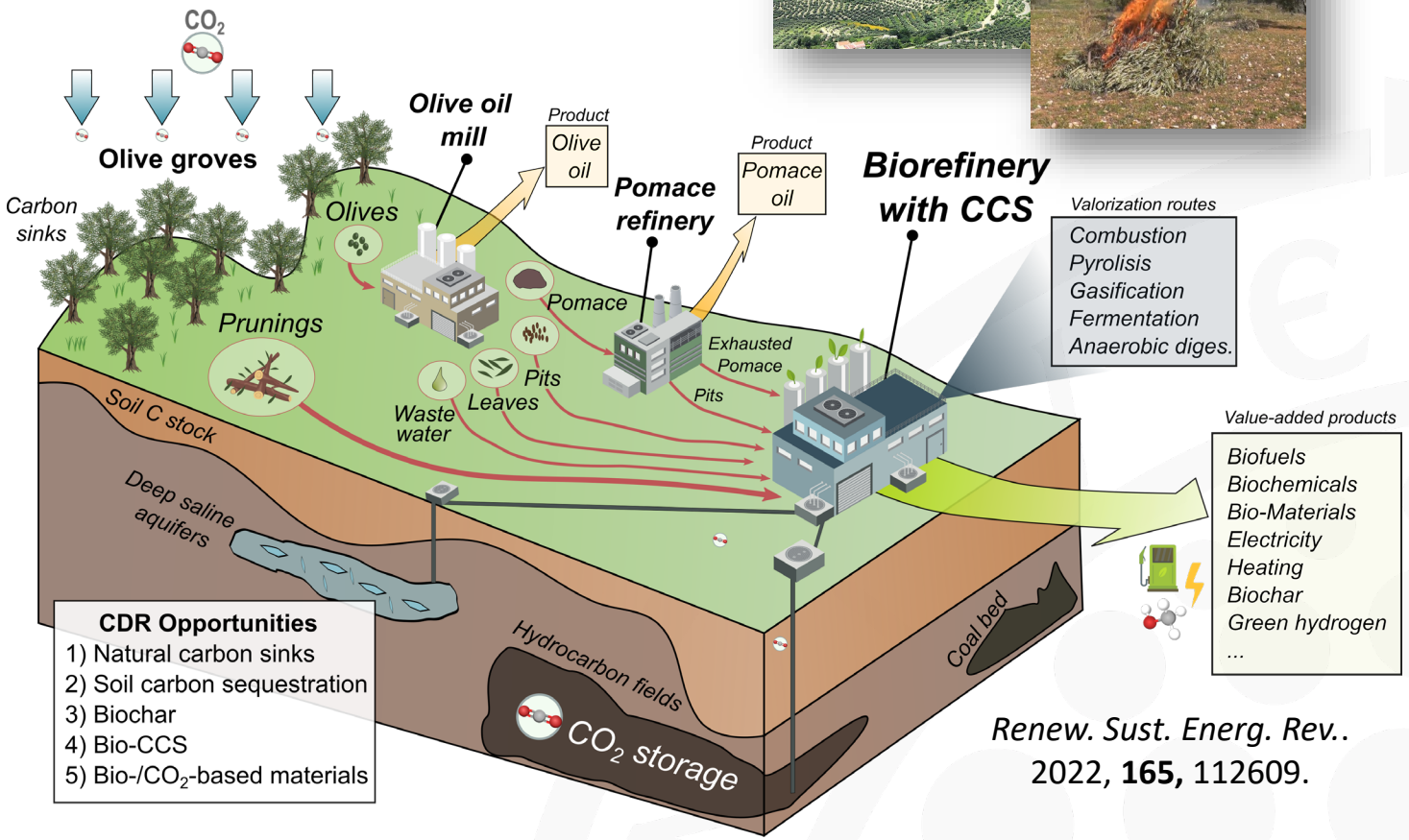


Renew. Sust. Energ. Rev.. 2022, **165,** 112609.

Olive-derived biomasses







Renew. Sust. Energ. Rev.. 2022, **165,** 112609.

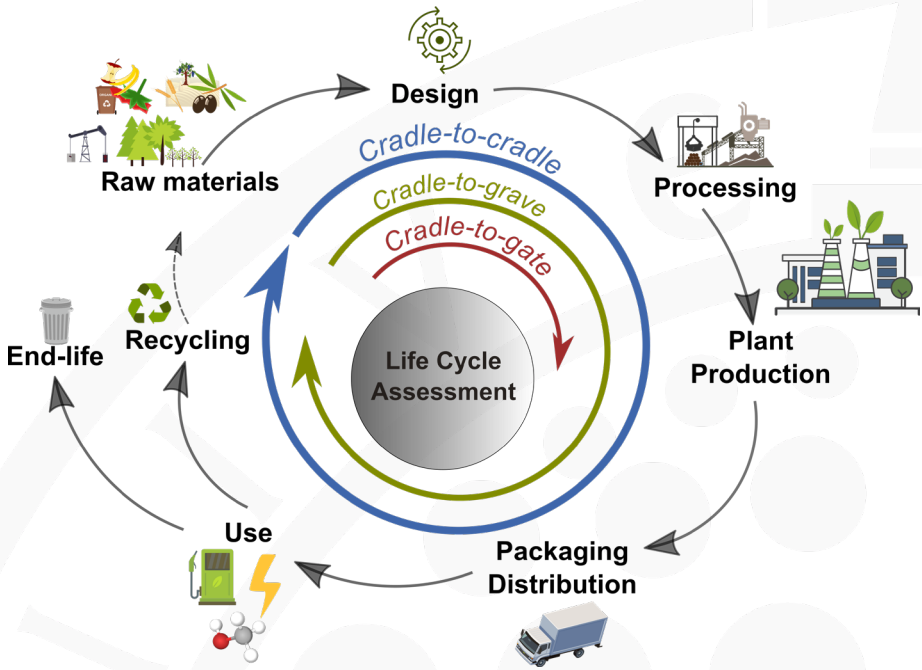
Case study: how integrating biorefineries w/wo CCS can valorize **exhausted olive pomace** — turning a local agro-industrial strategic residue into valuable bio-based products and energy.

Life Cycle Assessment

Life Cycle Assessment methodology

1. Goal and scope
2. Inventory analysis
3. Impact assessment
4. Interpretation

* following ISO 14040/44:2006 standards

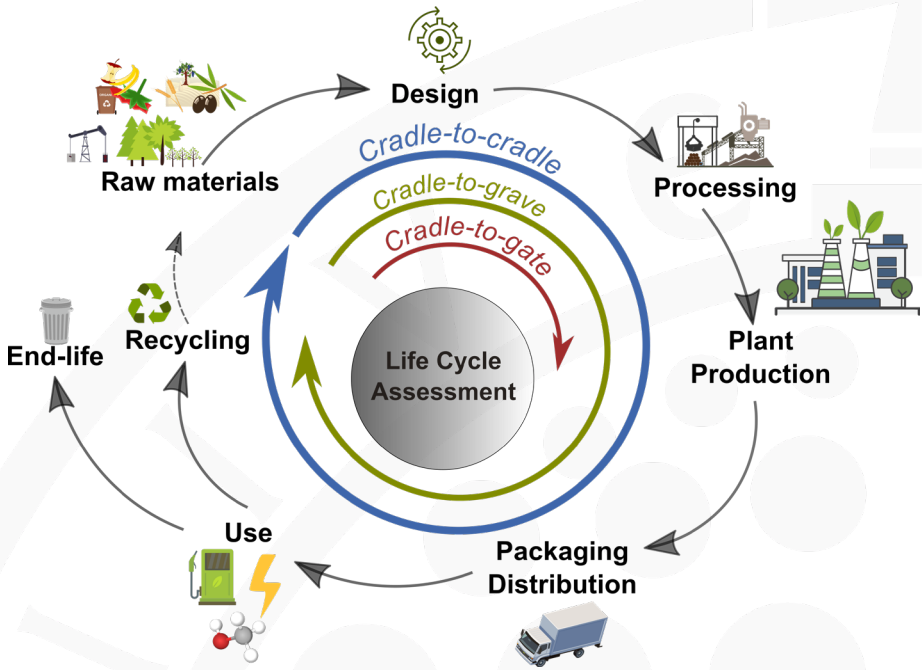


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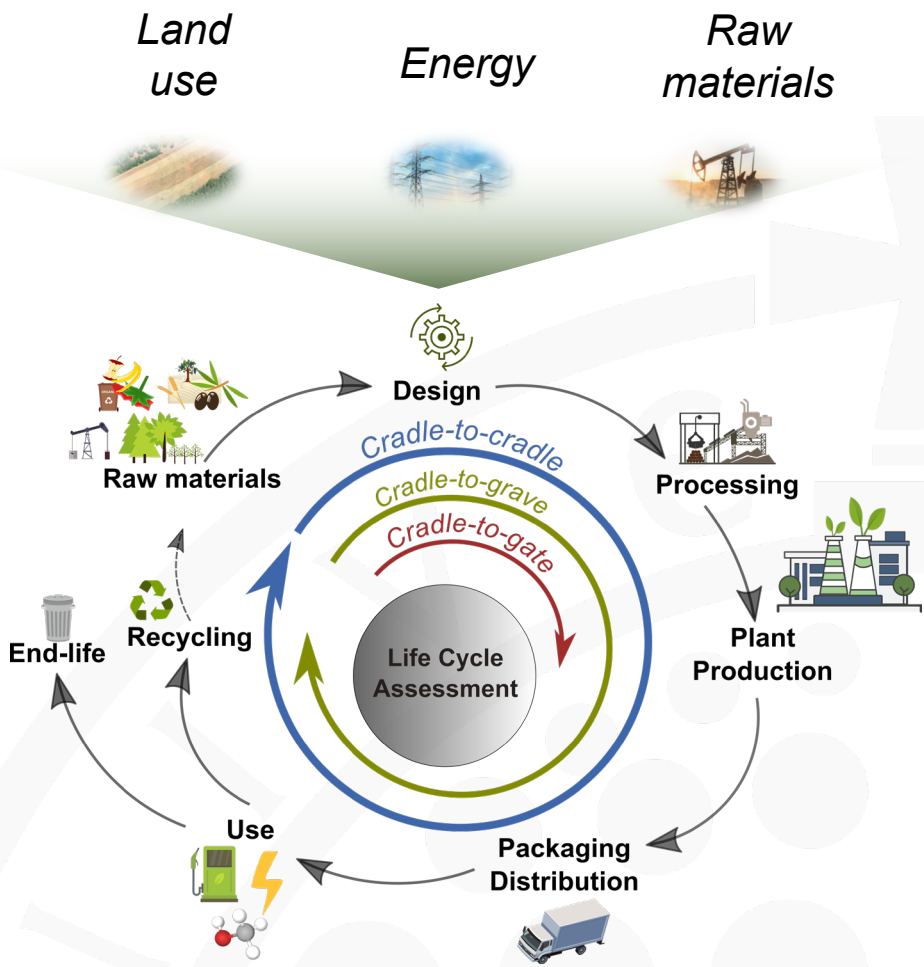


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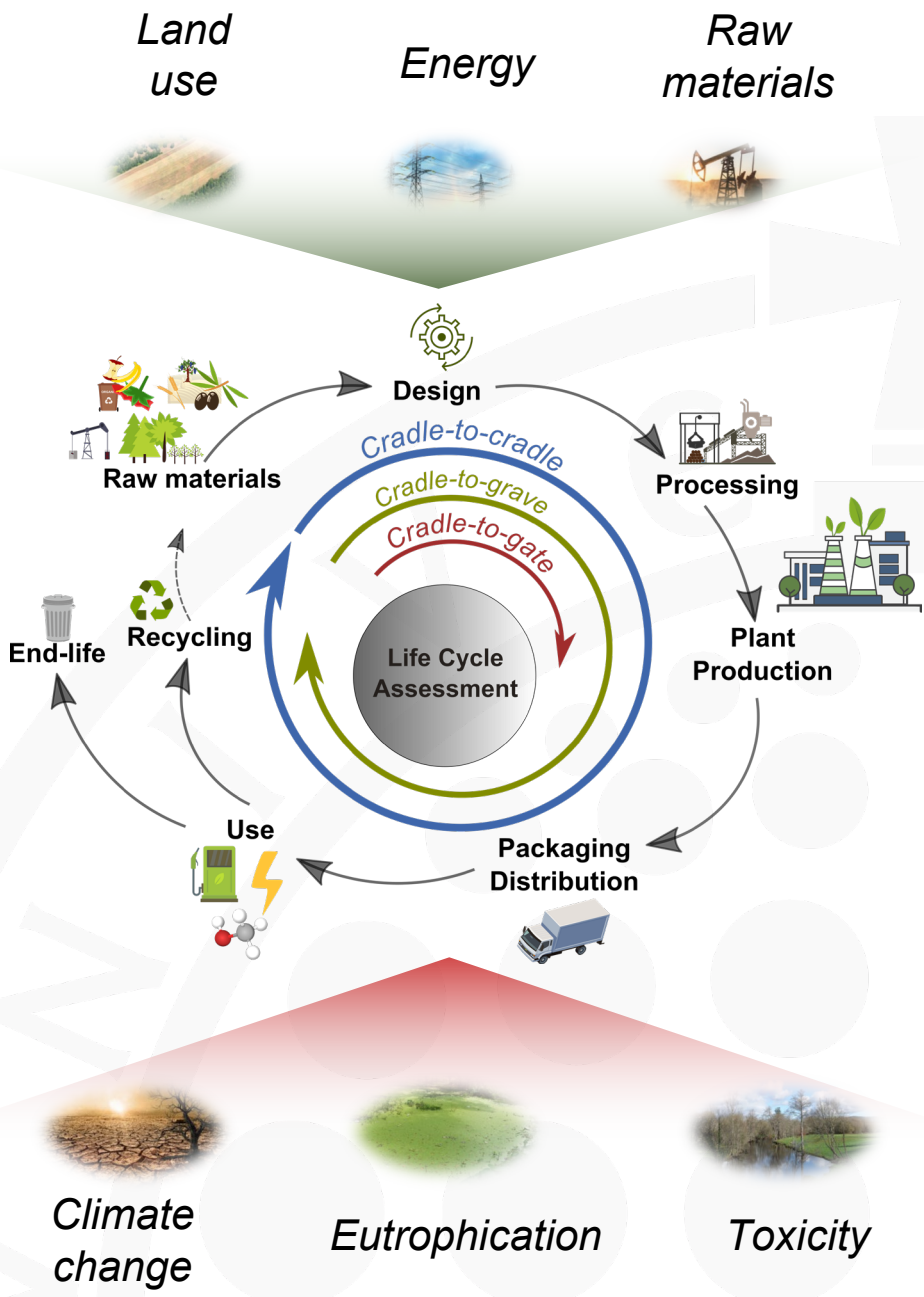


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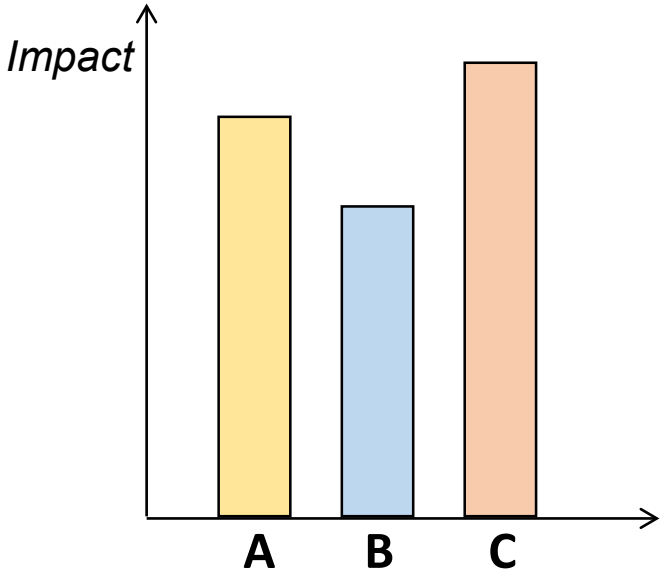


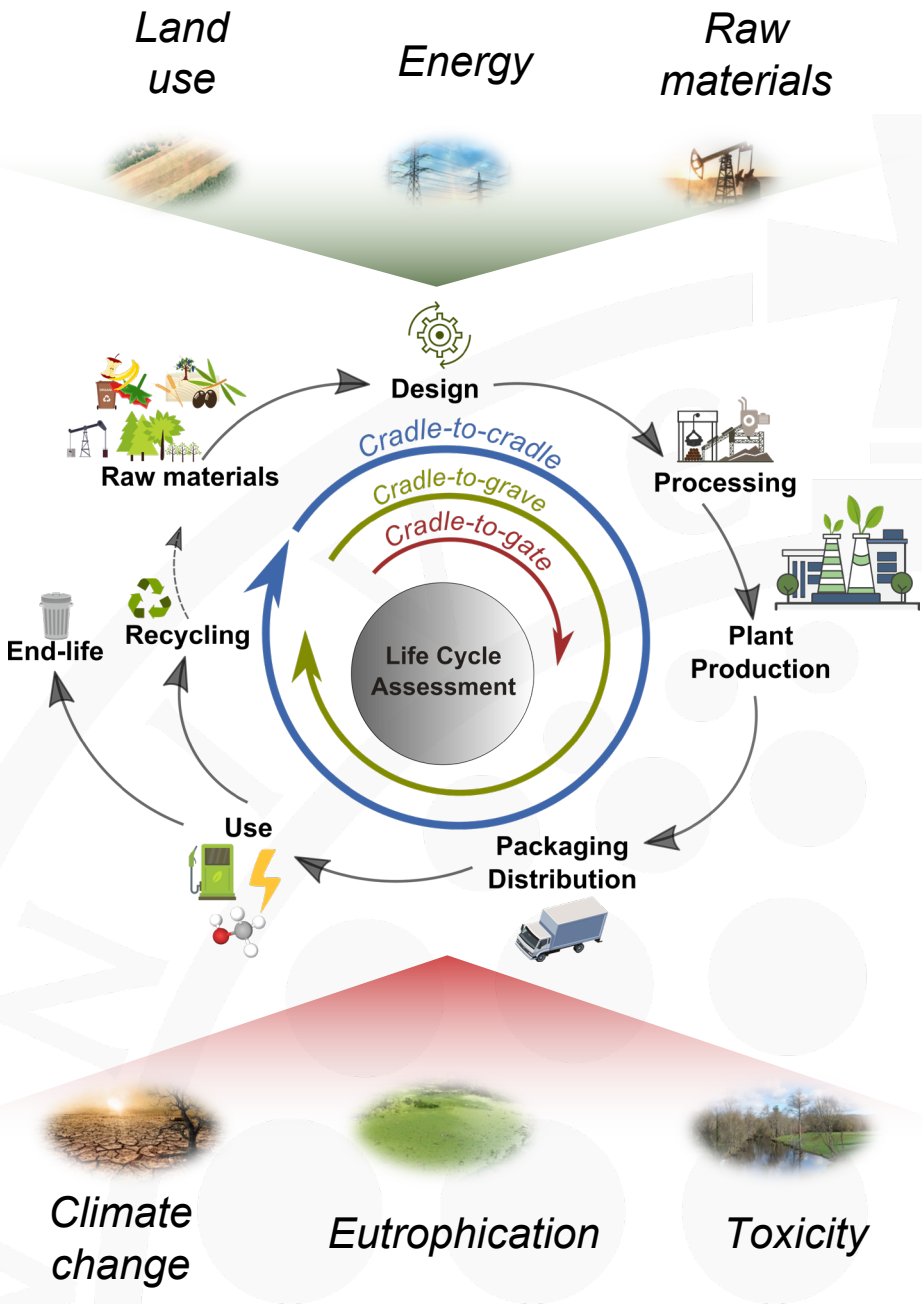
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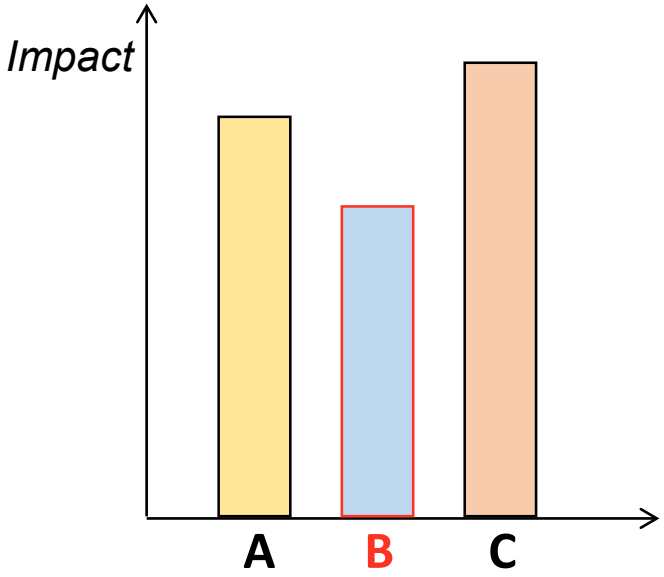


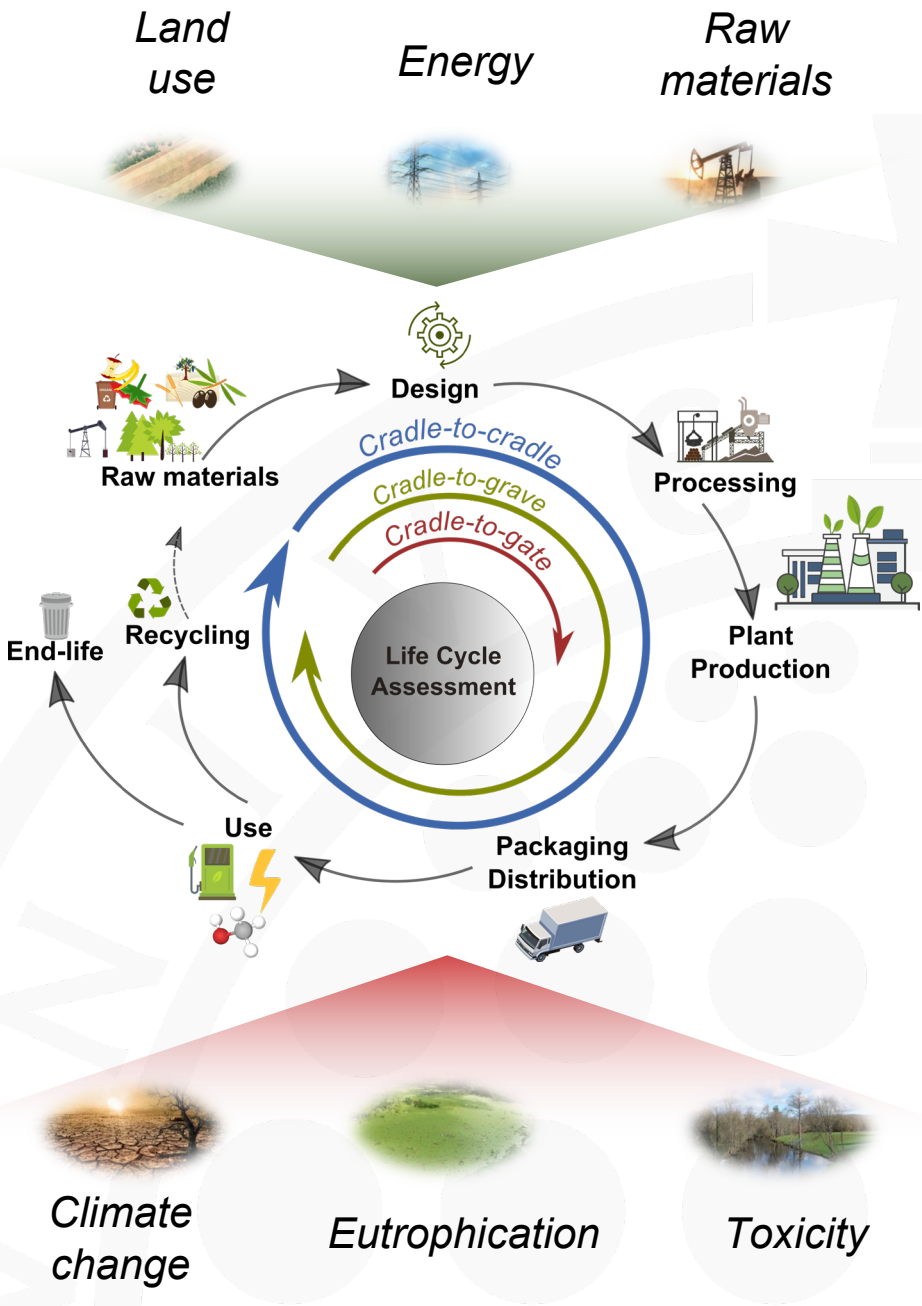
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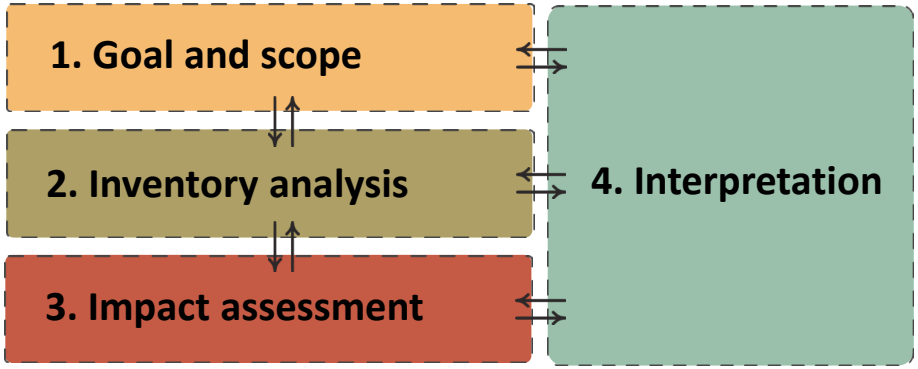
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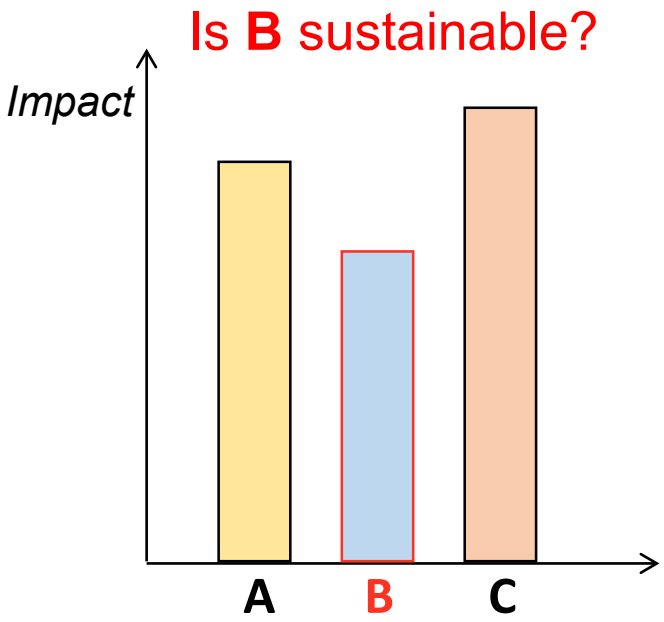


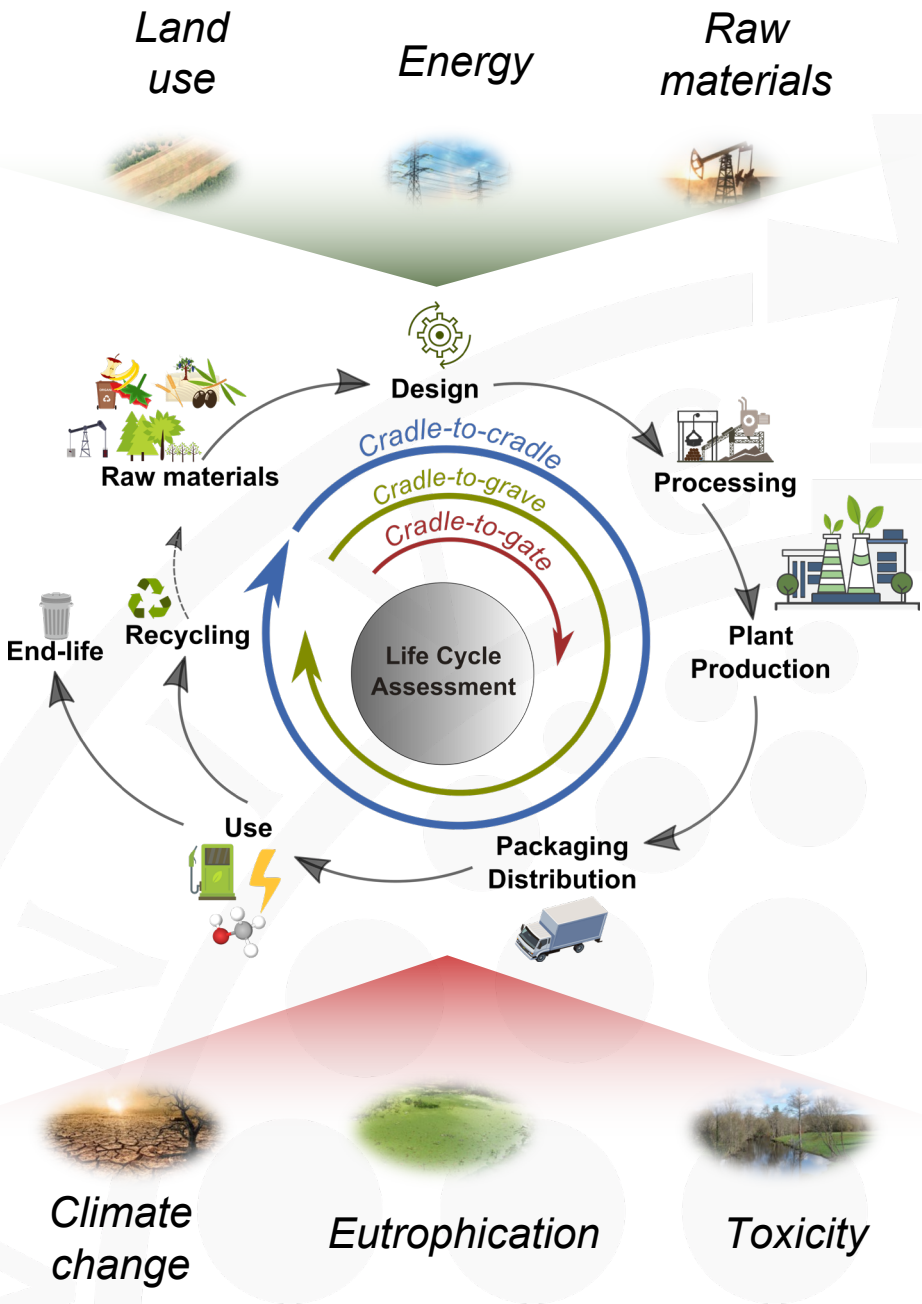
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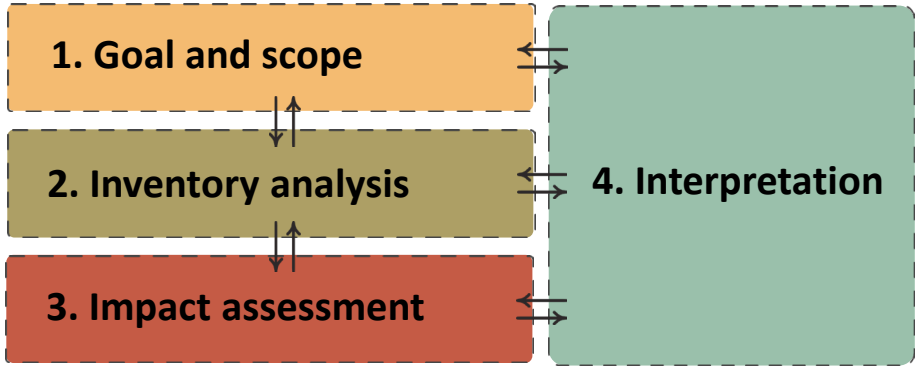
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Life Cycle Assessment

Life Cycle Assessment methodology

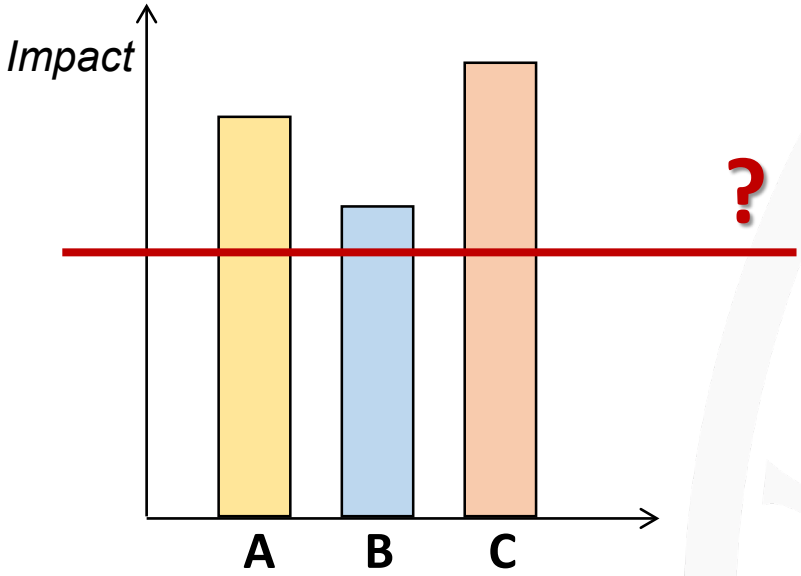


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How much

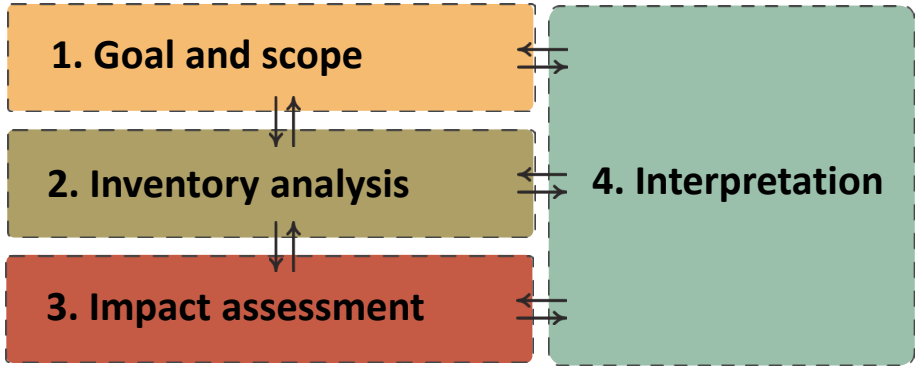
environmental impact

is too much?

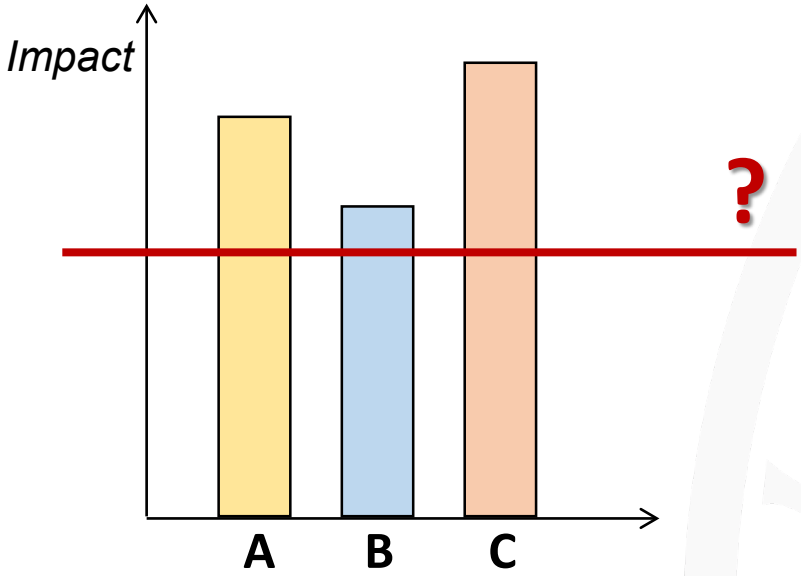


Life Cycle Assessment

Life Cycle Assessment methodology



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How much

environmental impact

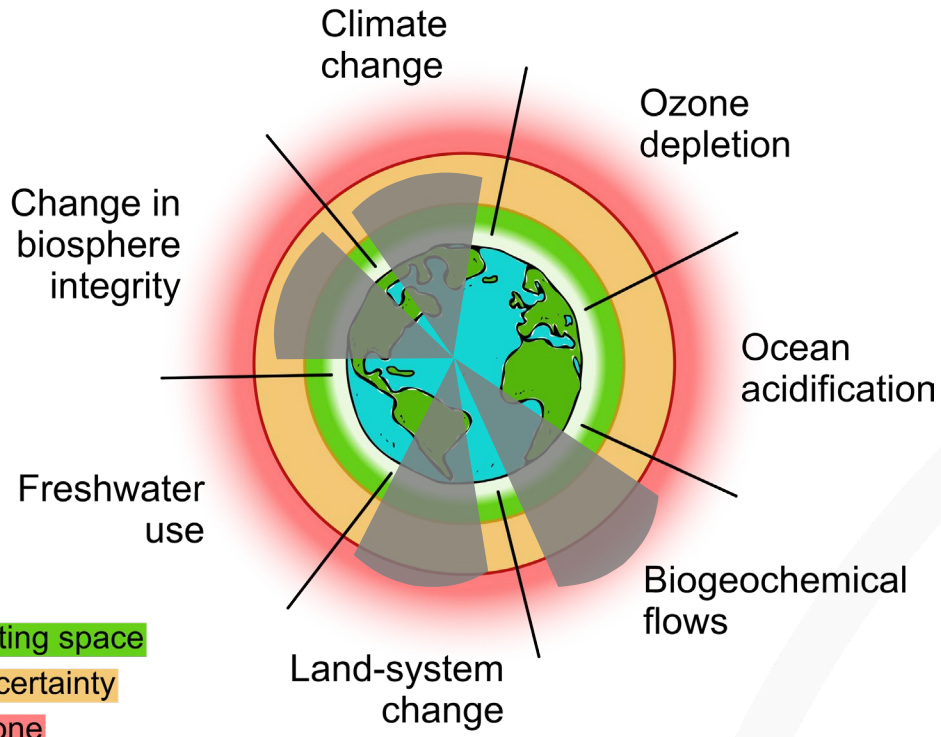
is too much?

development of

absolute sustainability

methods (**AESA**)

Absolute Sustainability Assessment



- ✓ **9 Earth-system processes**
- ✓ **7 quantified**
- ✓ **Safe Operating Space**
- ✓ **4** currently **transgressed**

Nature, 2009, **461**, 472–475

absolute sustainability

methods (**AESA**)

The **planetary boundaries** are biophysical **thresholds** that should **NOT** be exceeded to **operate safely**.

LCA-based AESA method

Ecological Indicators, 2018, **88**, 250–262